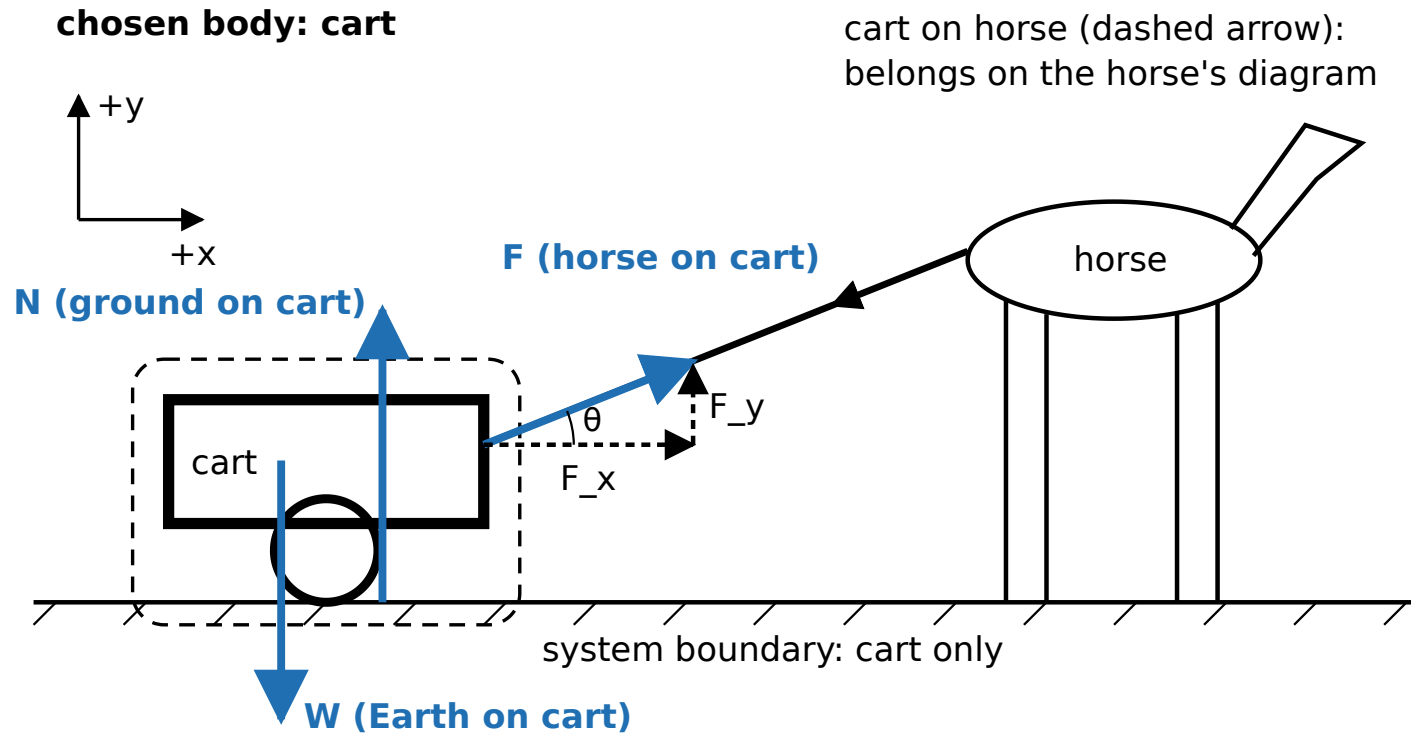


**Authored Grade-9 Force and Laws of Motion
spine covering balanced/unbalanced for:
Supported practice**

AUTHORED practice

Newton-law and observer-frame practice

A 4 kg body has a net external force of +12 N along a declared axis. Find its acceleration.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

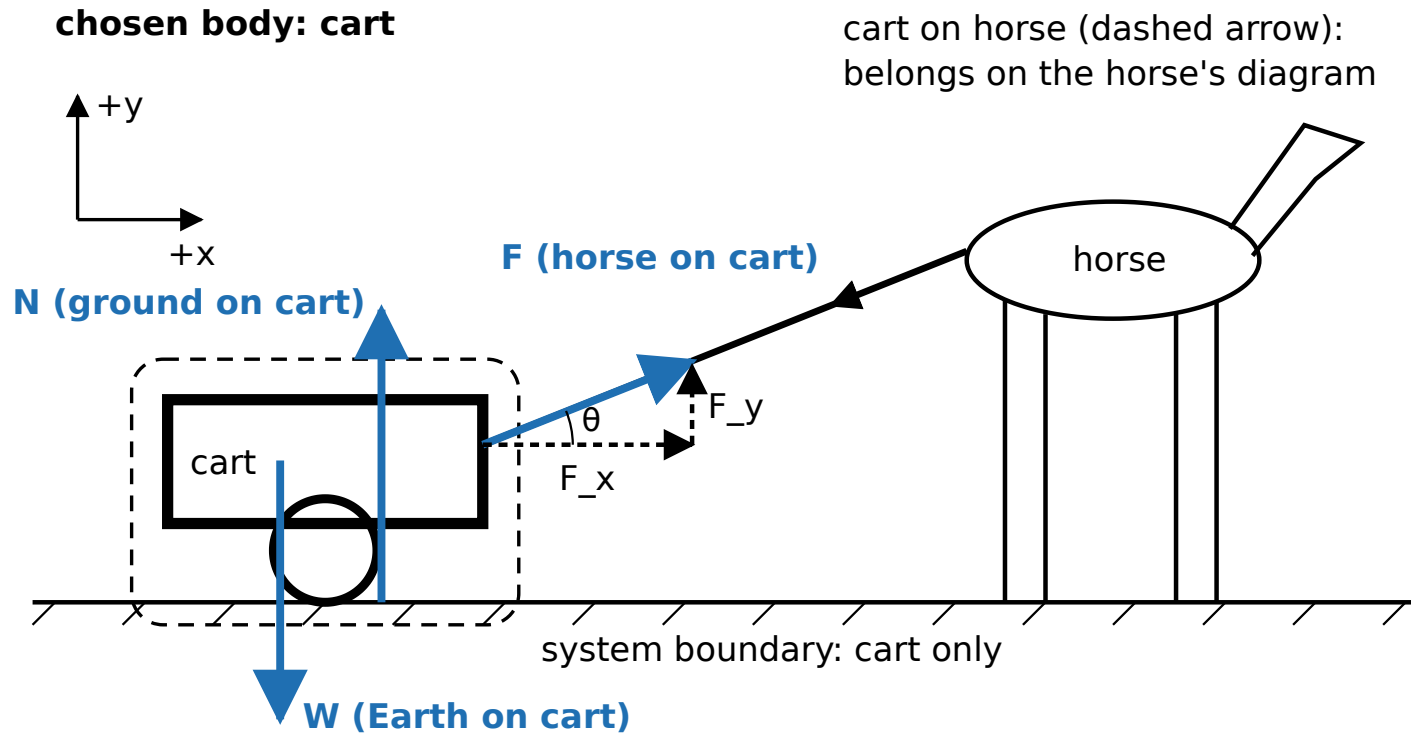
Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. Start by asking: Which body are we solving for?

Reasoning route and full solution



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. Use $F_{\text{net}} = m a$ for the chosen body.
2. $a = +12 \text{ N} / 4 \text{ kg} = +3 \text{ m/s}^2$.
3. The positive sign matches the net-force direction.

Answer

The acceleration is $+3 \text{ m/s}^2$.

Independent check

Multiplying 4 kg by $+3 \text{ m/s}^2$ recovers $+12 \text{ N}$.

If you went wrong

Your answer is 48 m/s^2 or $1/3 \text{ m/s}^2$: $F_{\text{net}} = m a$ means dividing the net force by the mass, $12/4$, not multiplying them or dividing the other way.

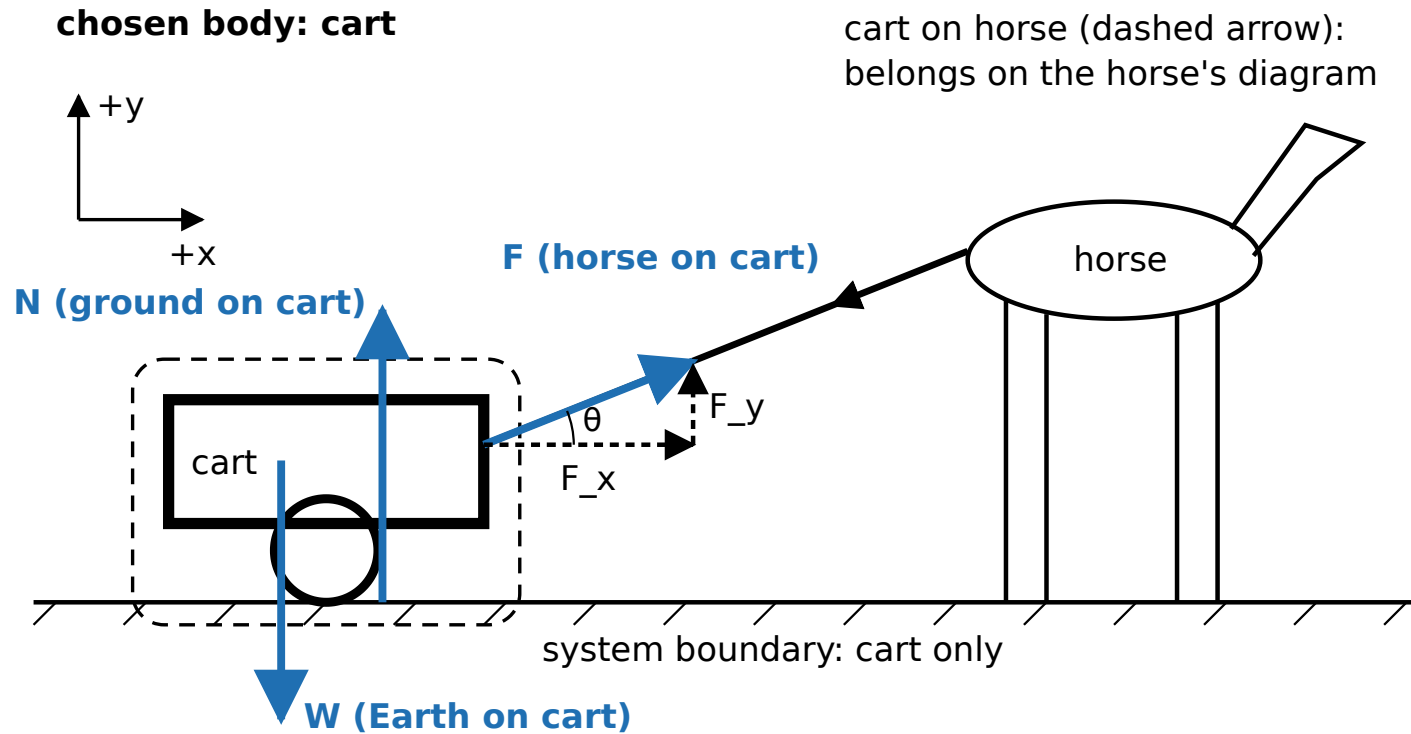
What this establishes

Established: with the net force known, $a = F_{\text{net}}/m$, pointing along the net force. The transfer task sets up the same law on an incline in two different choices of axes.

AUTHORED practice

Newton-law and observer-frame practice

One cart is at rest and another moves steadily in a straight line. For each, decide whether zero net force is compatible with the motion and explain the common rule.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

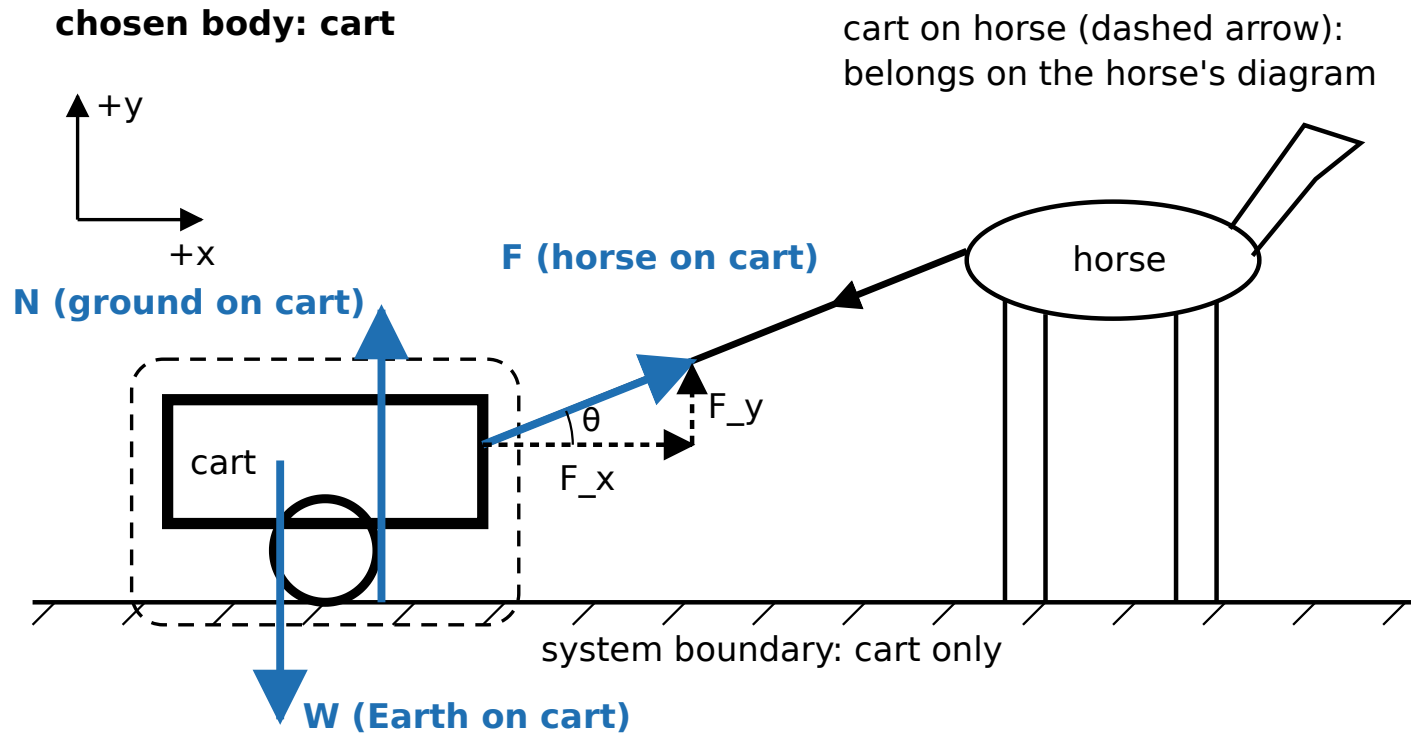
Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. Start by asking yourself: After the hand leaves the cart, is the hand still exerting the forward push?

Reasoning route and full solution



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. Rest is a zero velocity that stays zero.
2. Steady straight motion is a nonzero velocity that stays unchanged.
3. The shared condition is no change of velocity, not no motion.

Answer

Zero net force is compatible with both. Each cart can keep its current velocity unchanged.

Independent check

Ask whether the motion description contains speeding up, slowing down, or turning. If none occurs, zero net force remains compatible.

If you went wrong

You said the moving cart needs a forward force to keep going: that confuses having a velocity with changing it; steady straight motion needs zero net force, just as rest does.

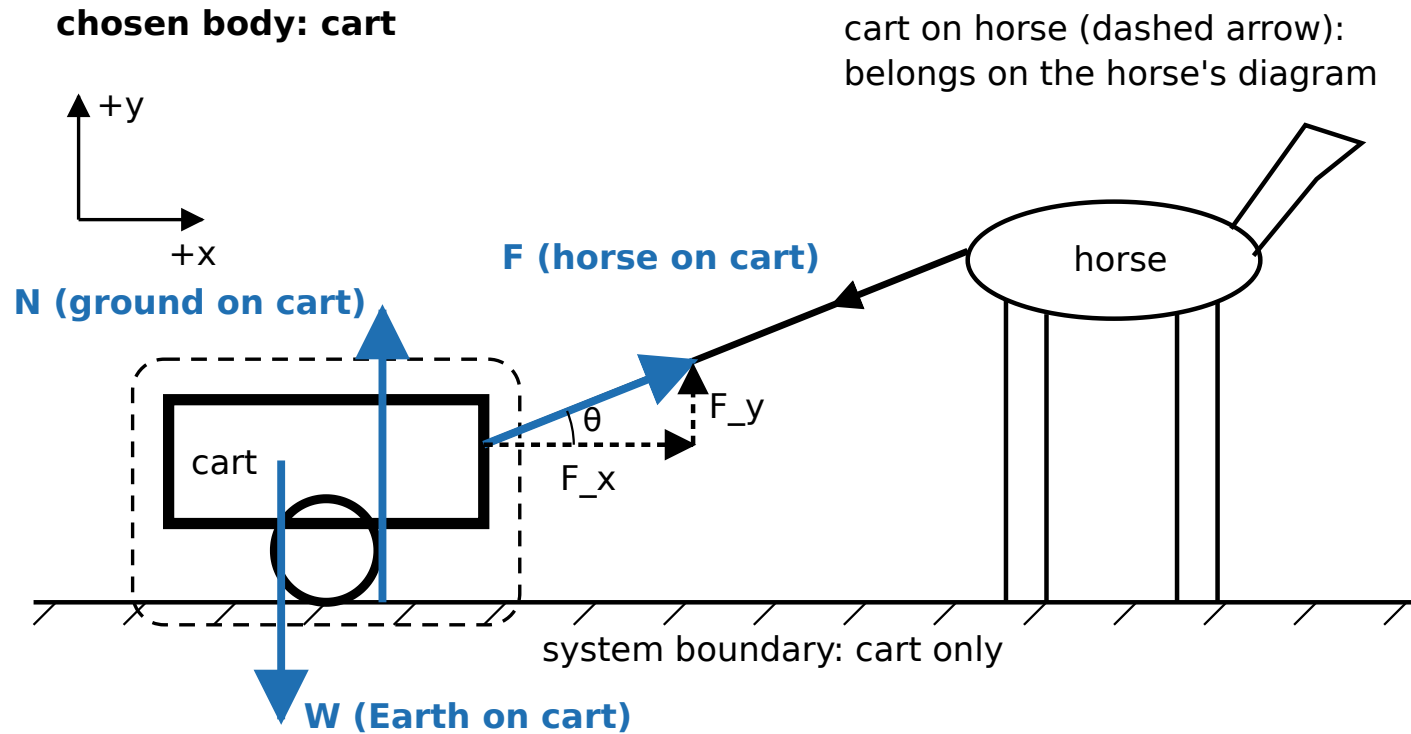
What this establishes

Established: zero net force means an unchanging velocity, and rest is the case where that velocity is zero. Later tasks read steady straight motion as a balanced-force state.

AUTHORED practice

Newton-law and observer-frame practice

An object has two equal opposite forces acting on it. Explain what is zero and what is not, then state what motion change the net result predicts.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

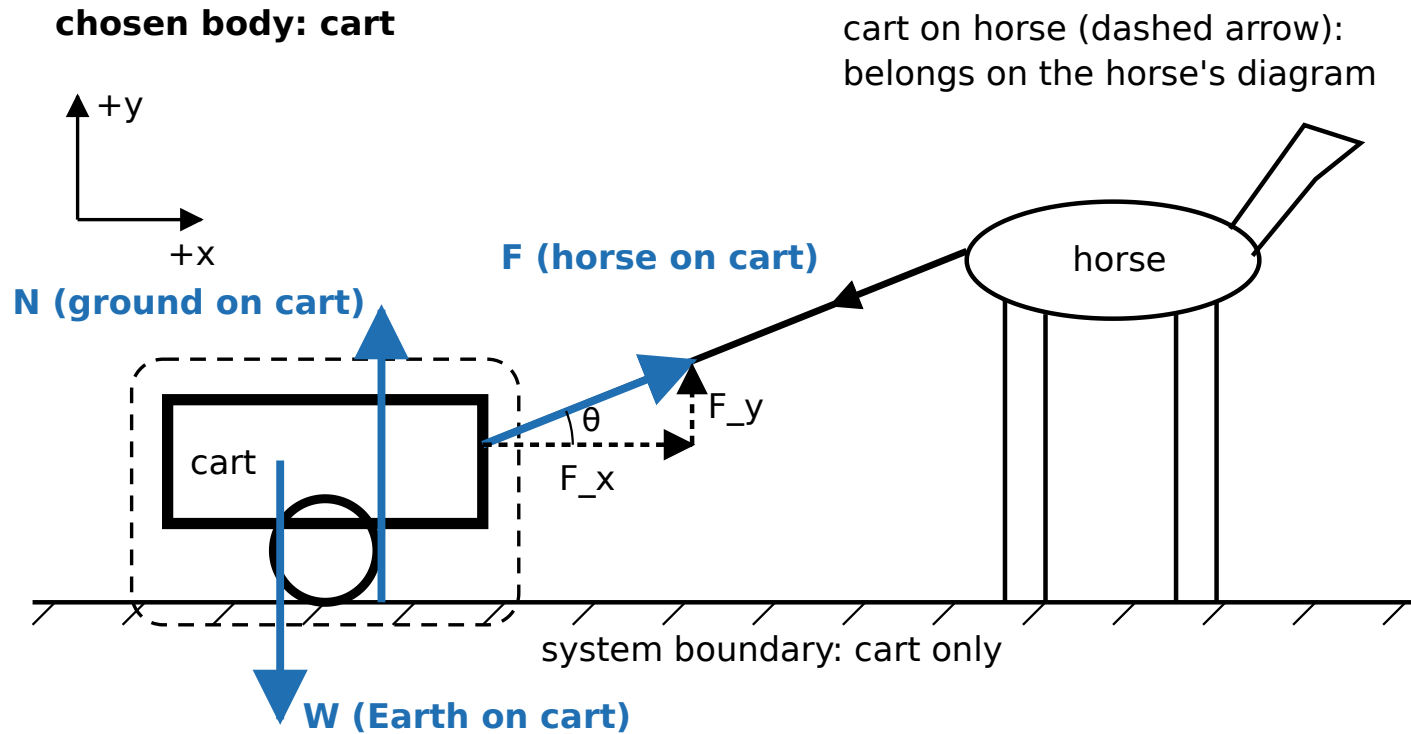
Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. Start by asking: Before adding anything, how many pushes are physically acting on the box?

Reasoning route and full solution



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. Each force remains an actual interaction on the body.
2. Opposite equal directional effects cancel only when combined.
3. A zero net force is consistent with unchanged velocity.

Answer

The net force is zero, but both individual forces are nonzero. The zero net result predicts no change of velocity.

Independent check

Increase both forces equally; the net remains zero even though neither individual force disappears.

If you went wrong

You wrote that no forces act: both forces are present and non-zero; only their sum is zero.

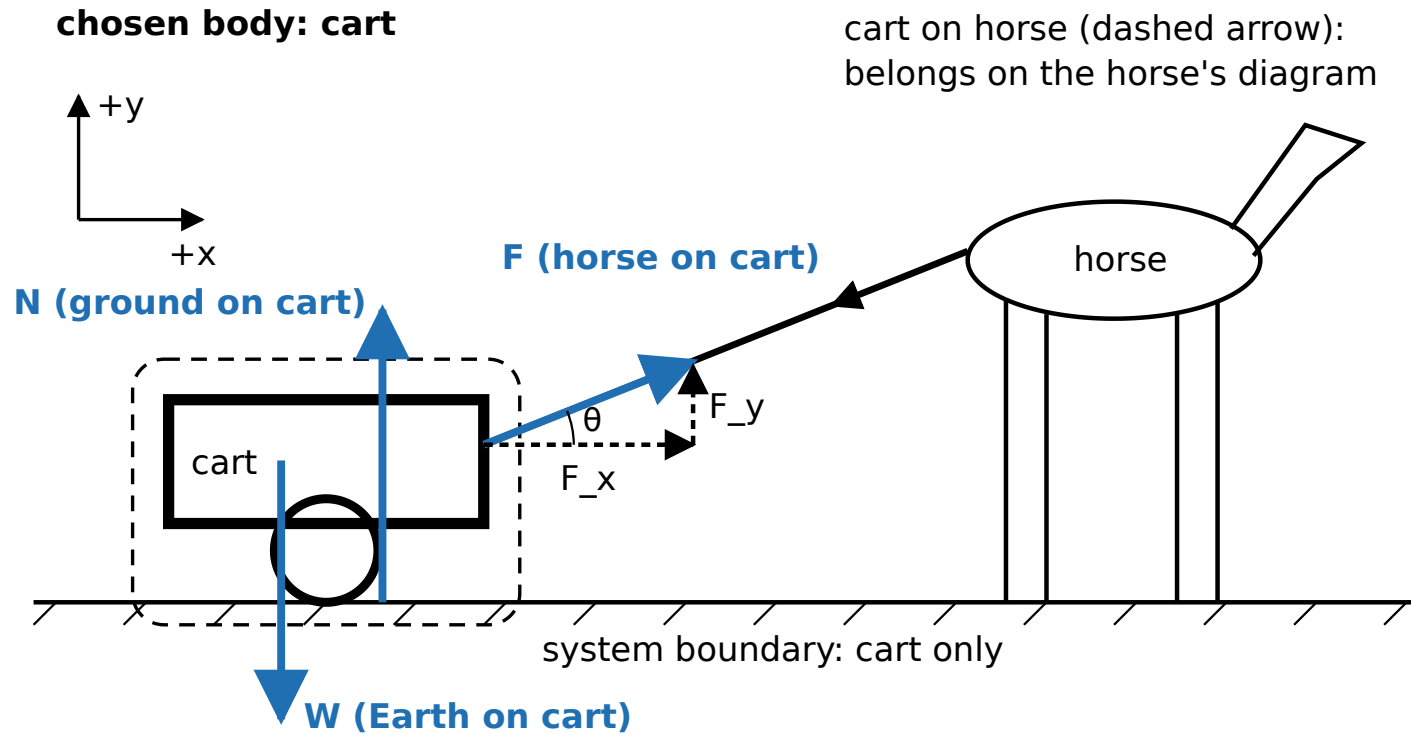
What this establishes

Established: forces can be real and non-zero while their sum is zero, and it is the sum that sets the change of velocity. Later tasks keep every force listed on a free-body diagram before any are combined.

AUTHORED practice

Newton-law and observer-frame practice

A person pushes a box. Describe the two contact-force partner arrows and state which one belongs on the box's free-body diagram.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

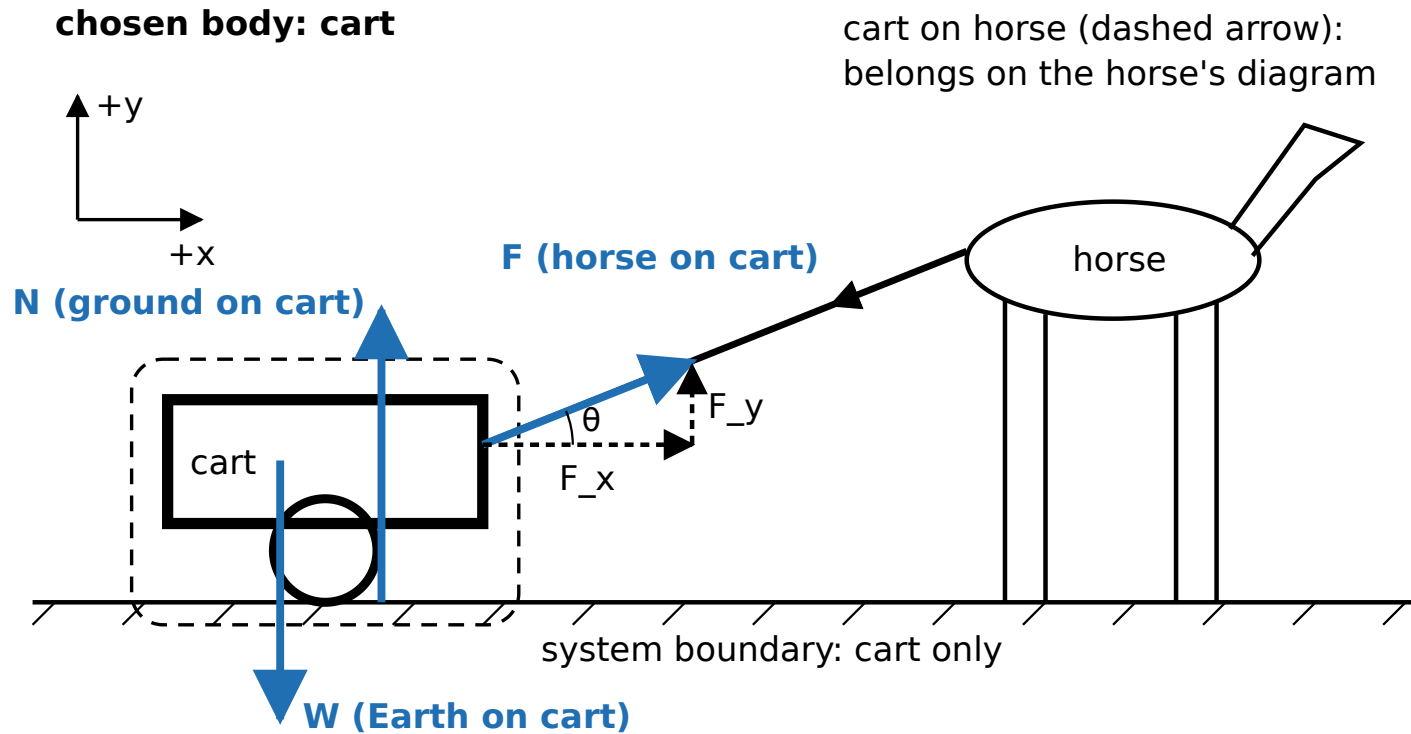
Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. Start by asking: Which single body are you describing first?

Reasoning route and full solution



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. Choose the box first.
2. Ask which member of the contact pair acts on the box.
3. The partner acts on the person, so it cannot cancel the first force on the box diagram.

Answer

The person's force on the box belongs on the box diagram. The box's equal-and-opposite force on the person belongs on the person's diagram.

Independent check

Switch the chosen body to the person; the membership of the two partner arrows must switch with it.

If you went wrong

You put both partner arrows on the box and let them cancel: the box's push on the person acts on the person, so it cannot be part of the box's sum.

What this establishes

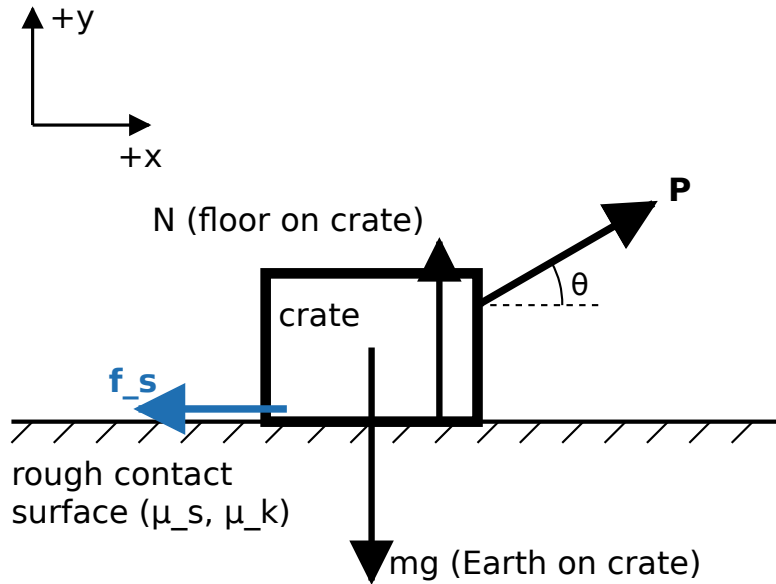
Established: a diagram holds only the forces on its own body, so a partner force never cancels on it.
The transfer task finds the same kind of double count hidden in resolved components on an incline.

AUTHORED practice

Newton-law and observer-frame practice

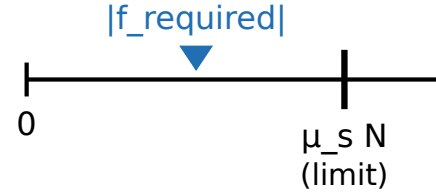
A box is placed on a conveyor belt moving to the right faster than the box. Before the box matches the belt speed, what direction is the friction force on the box?

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



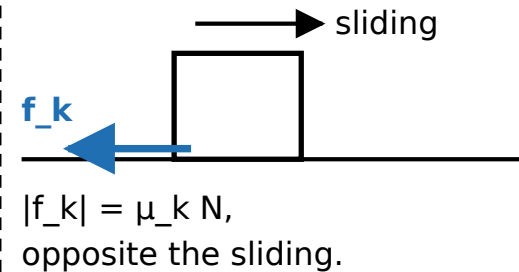
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



Stage 1 of 2

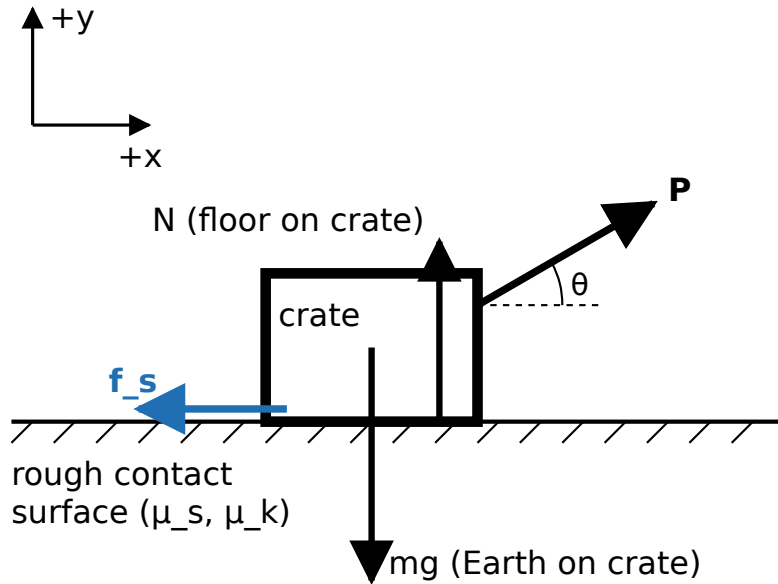
Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch to the

kinetic model only after sliding.

1. Start by asking: Which two surfaces are in contact?

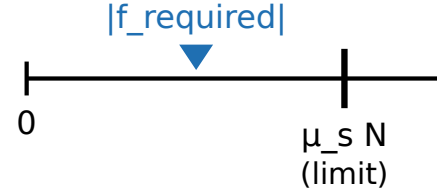
Reasoning route and full solution

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



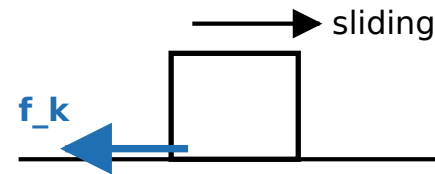
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Stage 1 of 5

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch

to the kinetic model only after sliding.

1. Relative to the belt, the box tends to slip left at the contact.
2. Friction opposes that relative slip tendency.
3. Therefore the belt's friction on the box points right and accelerates it.

Answer

The friction force on the box points to the right.

Independent check

The answer is allowed to point with the box's motion because friction is defined by the contact slip tendency, not by the box's velocity alone.

If you went wrong

Your friction arrow points left because the box 'lags behind': that is the way the box slips relative to the belt, and friction points the opposite way, to the right.

Repair

[Revisit: Imagine the relative sliding or tendency to slide if friction were absent, then point friction opposite that relative tendency.](#)

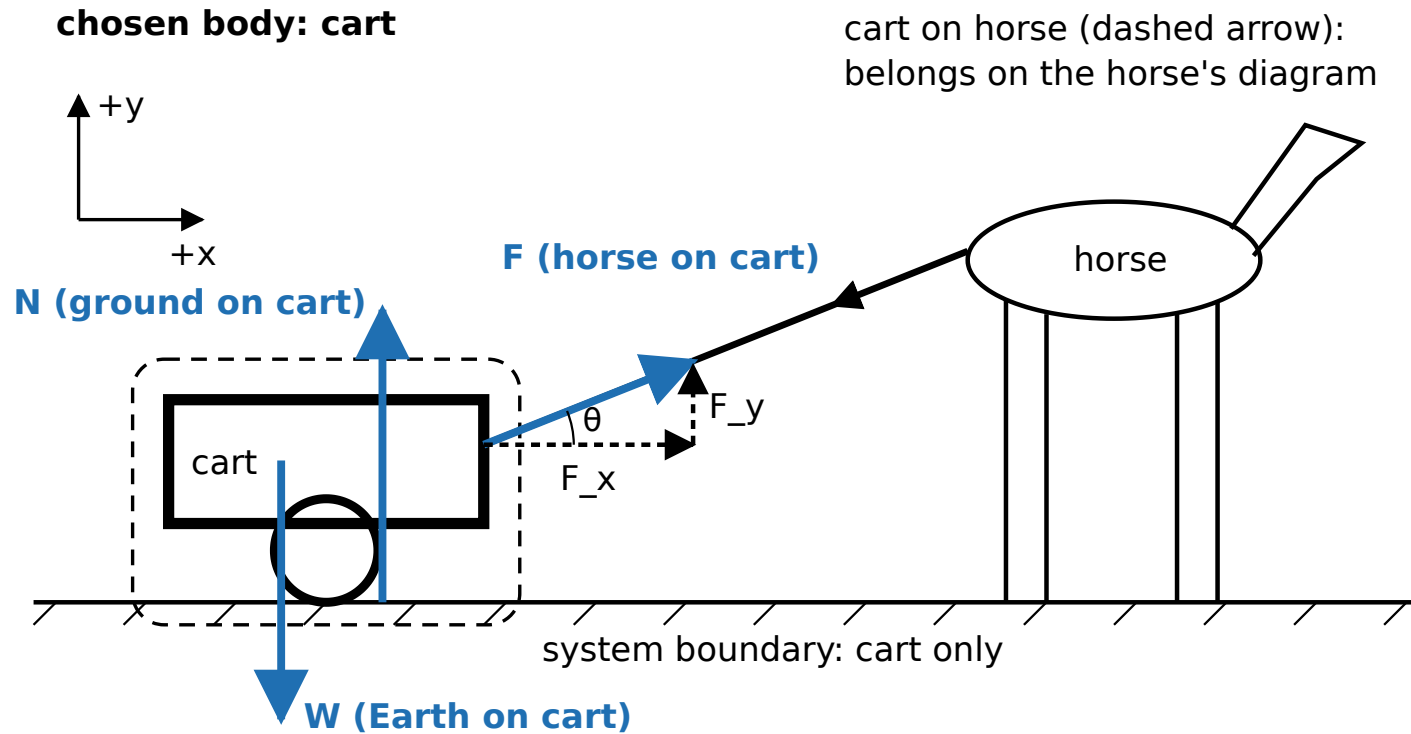
What this establishes

Established: friction's direction comes from the slip tendency at the contact, not from the object's velocity. The transfer task applies the same rule to a shoe pushing on the ground.

AUTHORED practice

Newton-law and observer-frame practice

A swimmer pushes water backward. Identify the third-law partner force and state which body each force acts on.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

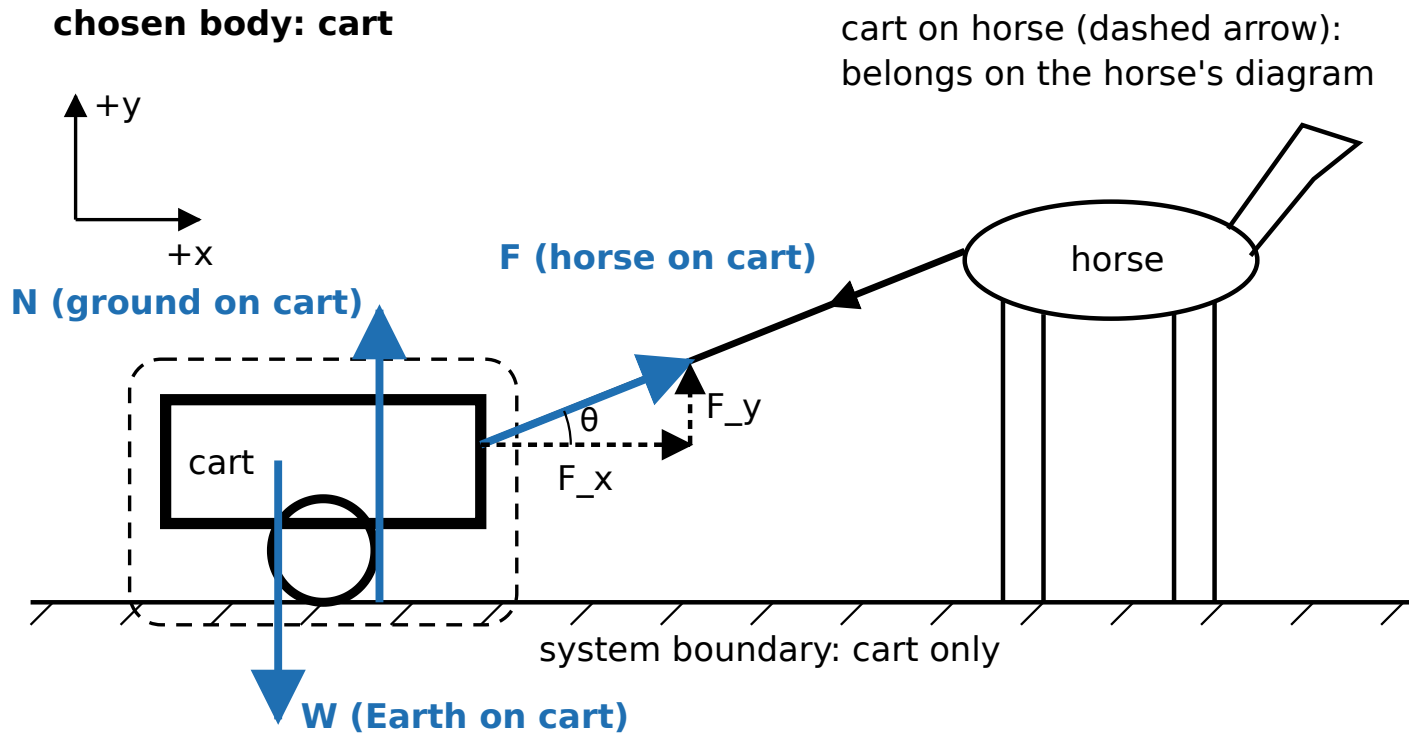
Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. Start by asking: What is the interaction?

Reasoning route and full solution



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. The interaction is swimmer-water contact.
2. Swimmer on water acts on the water.
3. Water on swimmer acts on the swimmer in the opposite direction.

Answer

The swimmer pushes the water backward, and the water pushes the swimmer forward with an equal-magnitude opposite force. The two forces act on different bodies.

Independent check

Only the water-on-swimmer force belongs on the swimmer's free-body diagram.

If you went wrong

You said the two forces cancel, so the swimmer cannot move: they act on different bodies, and the swimmer's own sum contains only the water's forward push.

What this establishes

Established: third-law partners are equal and opposite but act on different bodies, so each belongs to a different body's sum. Later tasks keep interaction pairs apart in connected-body problems.

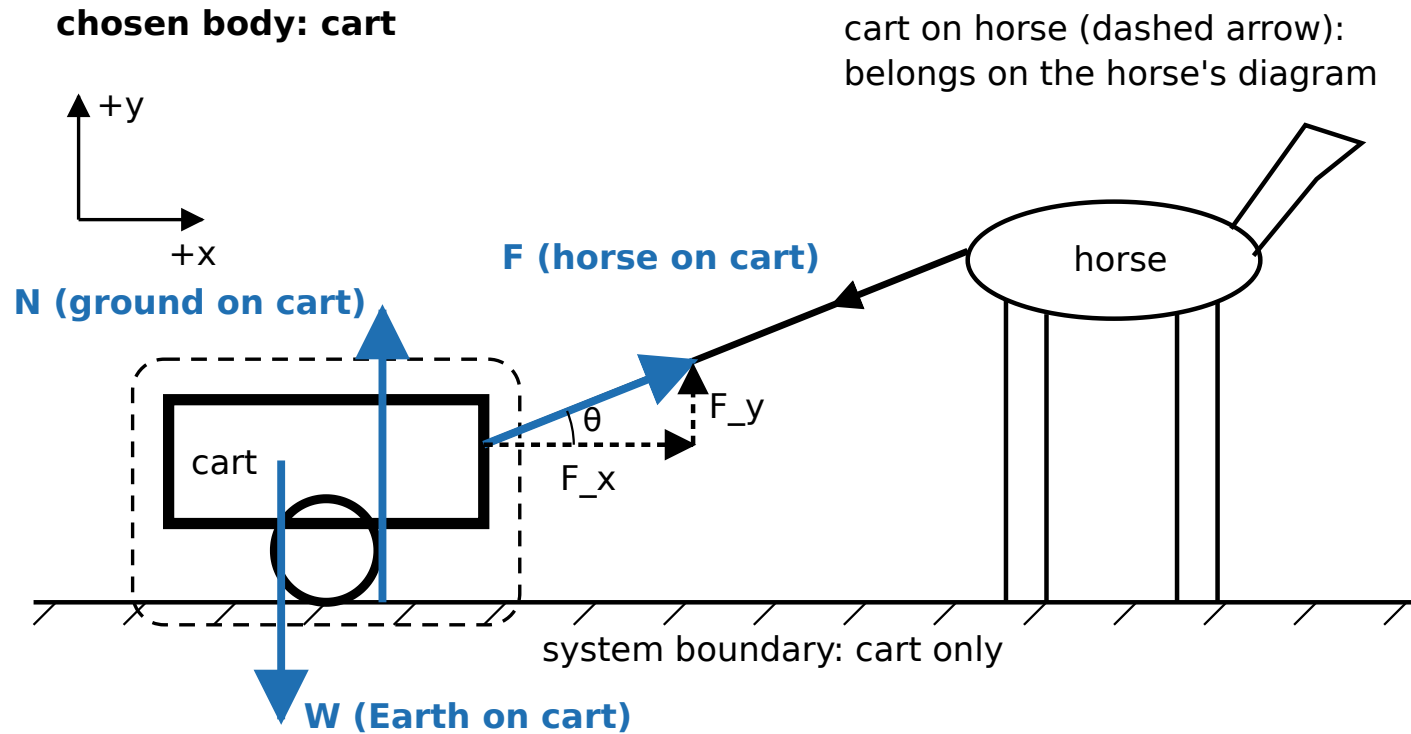
AUTHORED practice

Newton-law and observer-frame practice

Plan a trolley-pulley-hanging-mass experiment to verify Newton's second law. Describe one controlled variation that tests the effect of net force at fixed system mass, what you measure, and what result would support $F_{\text{net}} = m a$.

Conditions

- Treat friction as small or measure/account for it consistently.
- Use the same trolley-pulley system while changing the driving load.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

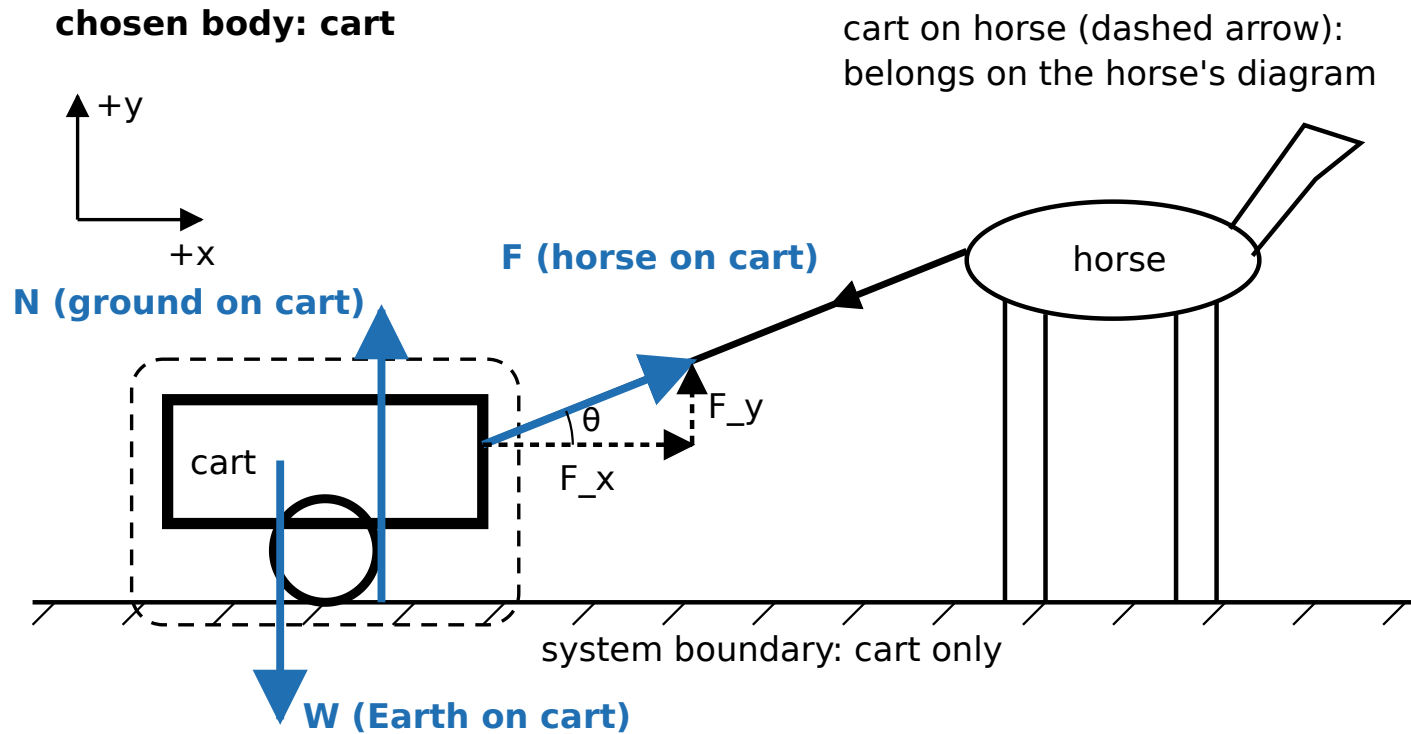
Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. To test the effect of force, do not let total moving mass become a second uncontrolled variable.

Reasoning route and full solution



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. Choose the trolley plus hanging mass as the moving system and keep its total mass fixed for the force-variation test.
2. Move a known mass from trolley to hanger to increase the driving weight without changing total moving mass.
3. Measure acceleration from distance/time data or a suitable motion sensor for each trial.
4. Correct for friction consistently if it is not negligible.
5. For fixed total mass, a plot of corrected F_{net} against a should be approximately linear through the origin within experimental uncertainty.

Answer

Keep the total moving mass approximately fixed while transferring masses between trolley and hanger so the driving force changes, measure acceleration for each trial, and test whether acceleration changes in direct proportion to the corrected net driving force.

Independent check

Changing force and total mass at the same time without controlling either would not isolate the F_{net} - a relationship.

If you went wrong

Your plan adds new masses to the hanger from outside: then the driving force and the moving mass both change, so a change in acceleration cannot be put down to force alone.

What this establishes

Established: to test $F_{\text{net}} = m a$, vary the net driving force while the system mass stays fixed, by moving masses between trolley and hanger. Later tasks use the same system boundary to find the acceleration of connected bodies.

AUTHORED practice

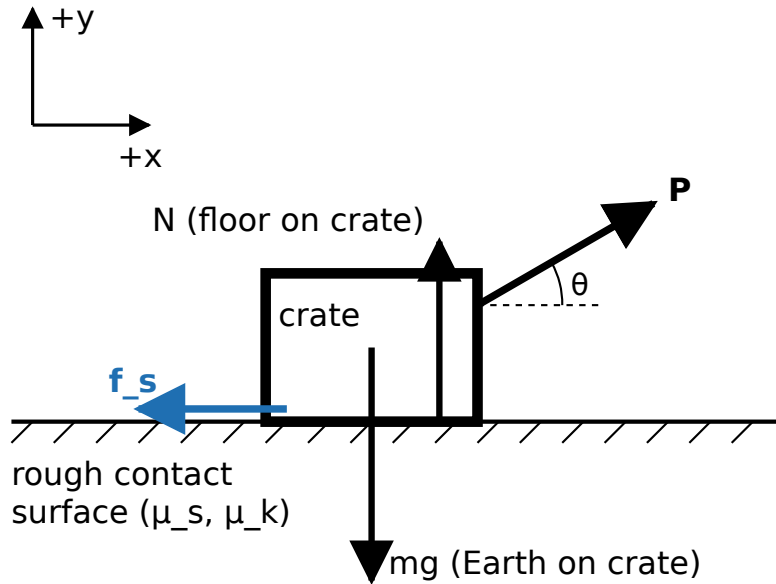
Newton-law and observer-frame practice

A small block of mass m rests on a larger block of mass M . The floor under the larger block is smooth. A horizontal force F is applied to the larger block. The coefficient of static friction between the blocks is μ_s . Derive the largest value of F for which the two blocks can move together without slipping.

Conditions

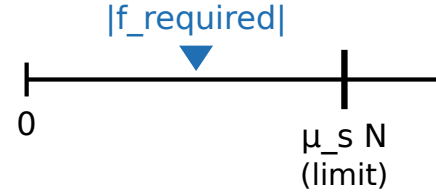
- The upper block has no other horizontal force.
- The contact surfaces remain horizontal.
- Use the static-friction limit only at impending slip.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



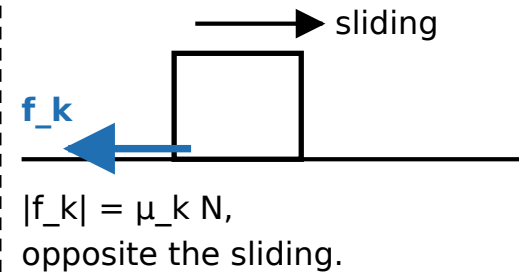
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



Stage 1 of 2

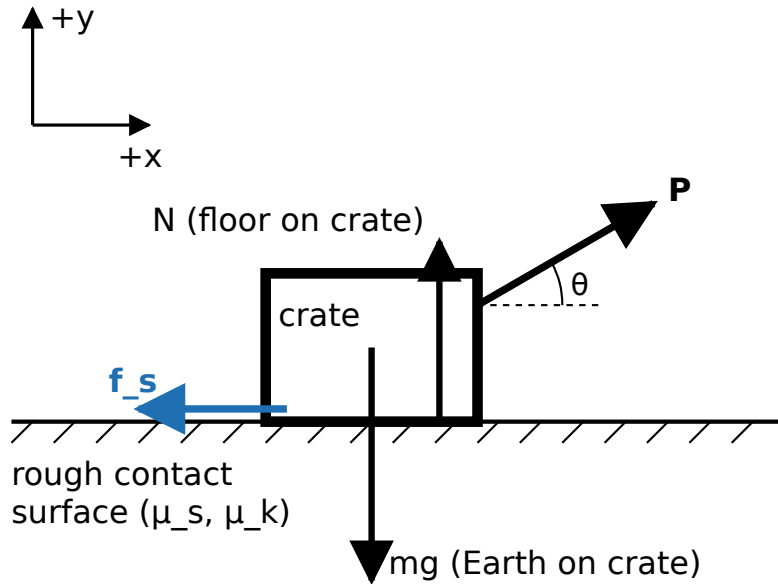
Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch to the

kinetic model only after sliding.

1. First suppose the blocks do not slip. What acceleration would the two-block system then have?

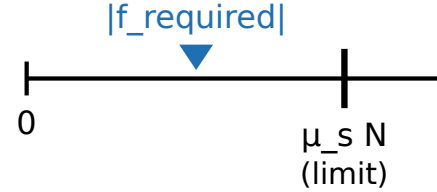
Reasoning route and full solution

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



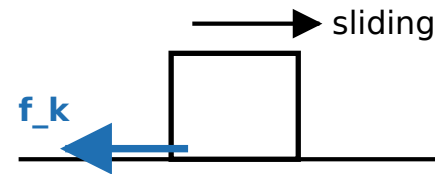
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Stage 1 of 5

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch

to the kinetic model only after sliding.

1. Treat both blocks together first: the only external horizontal force on the two-block system is F , so $a = F/(M+m)$.
2. For the upper block alone, static friction is the horizontal force that must provide acceleration a , so $f_{\text{required}} = m a = mF/(M+m)$.
3. Because the contact is horizontal with no other vertical force on the upper block, $N = m g$.
4. Static contact is possible only while $f_{\text{required}} \leq \mu_s N$; equality occurs at impending slip.
5. Substituting and cancelling m gives $F_{\text{max}} = \mu_s(M+m)g$.

Answer

While there is no slip, both blocks have $a = F/(M+m)$. The upper block needs friction $f_{\text{required}} = m a$. Since $N = m g$ here, no slip requires $mF/(M+m) \leq \mu_s m g$, so $F \leq \mu_s(M+m)g$. Therefore $F_{\text{max}} = \mu_s(M+m)g$.

Independent check

For F below this threshold, the required friction is below its maximum; setting $f_s = \mu_s N$ for every smaller F would incorrectly predict the same friction at all forces.

If you went wrong

Your answer is $F_{\text{max}} = \mu_s m g$: that is the friction limit itself, but F must accelerate both blocks, so the limit is scaled up by $(M + m)/m$.

Repair

[Revisit: Compare \$|f_{\text{required}}|\$ with \$\mu_s N\$.](#)

What this establishes

Established: for a no-slip threshold, find the shared acceleration from the whole system, then test the friction one block needs against $\mu_s N$. The transfer task leaves the contact state open, so you must first decide whether rest can hold at all.

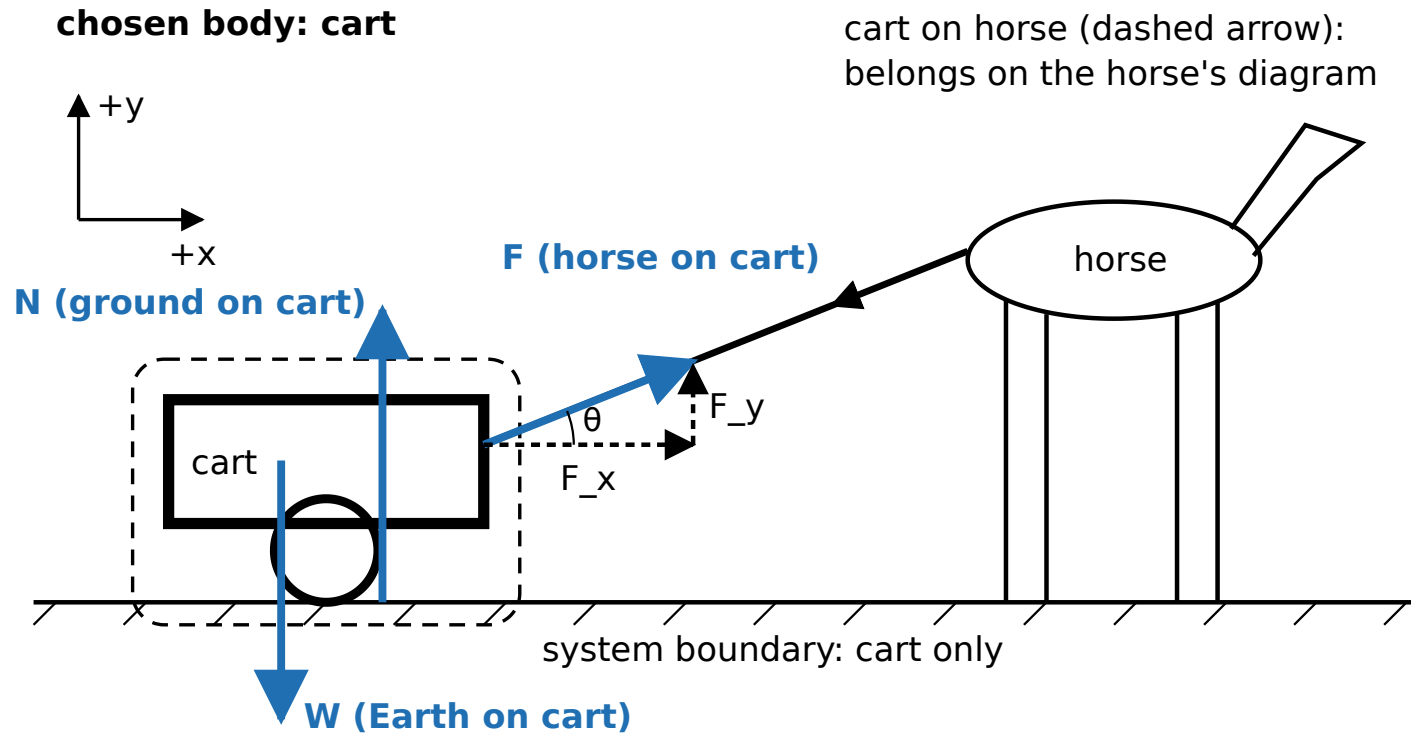
AUTHORED practice

Newton-law and observer-frame practice

Two blocks A and B of masses m_A and m_B remain in contact on a smooth horizontal floor. A horizontal force F pushes A toward B, and the contact is maintained. Derive their common acceleration and the magnitude of the contact force exerted by A on B.

Conditions

- The floor is frictionless.
- The blocks remain in contact throughout the motion.
- Take the direction of F as positive.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

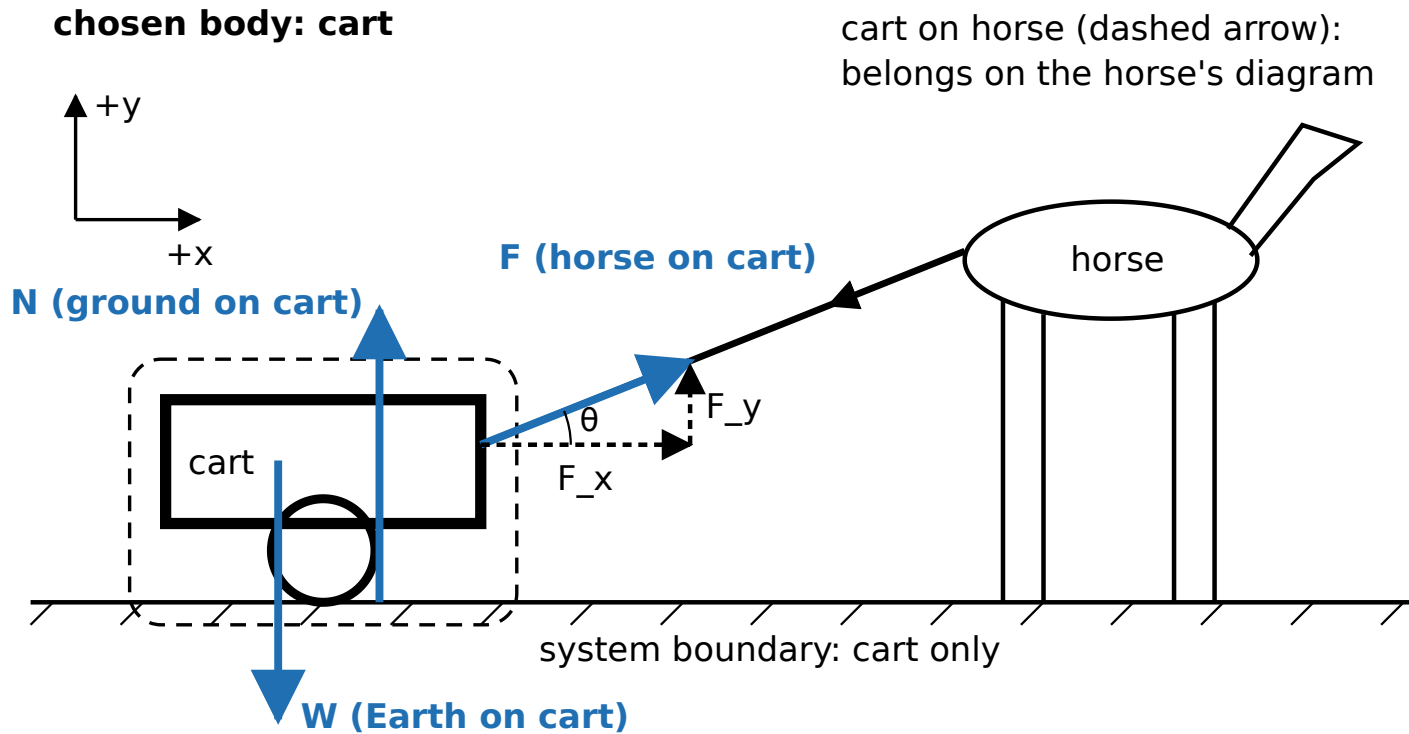
Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. What physical condition lets you use one acceleration for both blocks?

Reasoning route and full solution



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. Because the maintained contact fixes the separation of A and B along the motion direction, they share one acceleration component.
2. For the combined system A+B, the contact pair is internal and the external horizontal force is F , giving $a = F/(m_A + m_B)$.
3. For B alone, the force exerted by A is external to B and must remain on B's free-body diagram.
4. Newton II for B gives $N_{AB} = m_B a = m_B F/(m_A + m_B)$.

Answer

The fixed contact keeps the blocks at one common acceleration $a = F/(m_A + m_B)$. For block B alone, the contact force is its only horizontal force, so $N_{AB} = m_B a = m_B F/(m_A + m_B)$.

Independent check

If m_B tends to zero, the required contact force tends to zero while the system acceleration tends to F/m_A , which is consistent.

If you went wrong

Your contact force equals F : the push on A is not passed to B undiminished, because part of F accelerates A itself; B needs only $m_B a$.

Repair

Revisit: Match the chosen system boundary to the requested quantity: use the combined constrained system when internal forces are irrelevant to the shared acceleration, then isolate the body or subsystem that makes a requested interaction force external.

What this establishes

Established: maintained contact gives one acceleration, the whole system finds it, and the body that feels a contact force is isolated to find that force. The transfer task adds a third cart and leaves the choice of system to you.

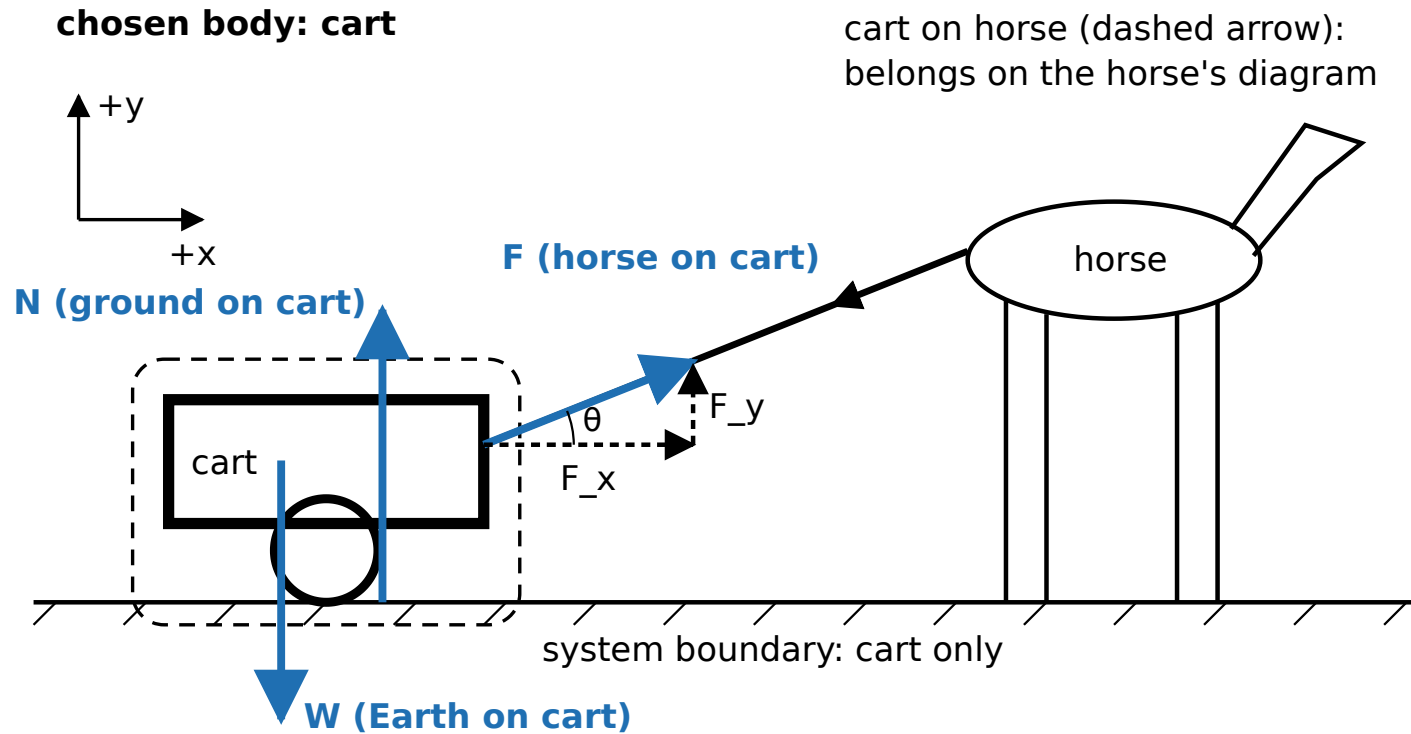
AUTHORED practice

Newton-law and observer-frame practice

Two carts A and B of masses m_A and m_B are connected by one taut massless inextensible string on a smooth horizontal track. A horizontal force F pulls cart A away from B along the string. Derive the common acceleration and the string tension. Then name one model change that would make equal tension throughout the connector no longer guaranteed.

Conditions

- The string is massless, taut and inextensible.
- There is no pulley in this question.
- The track is frictionless.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

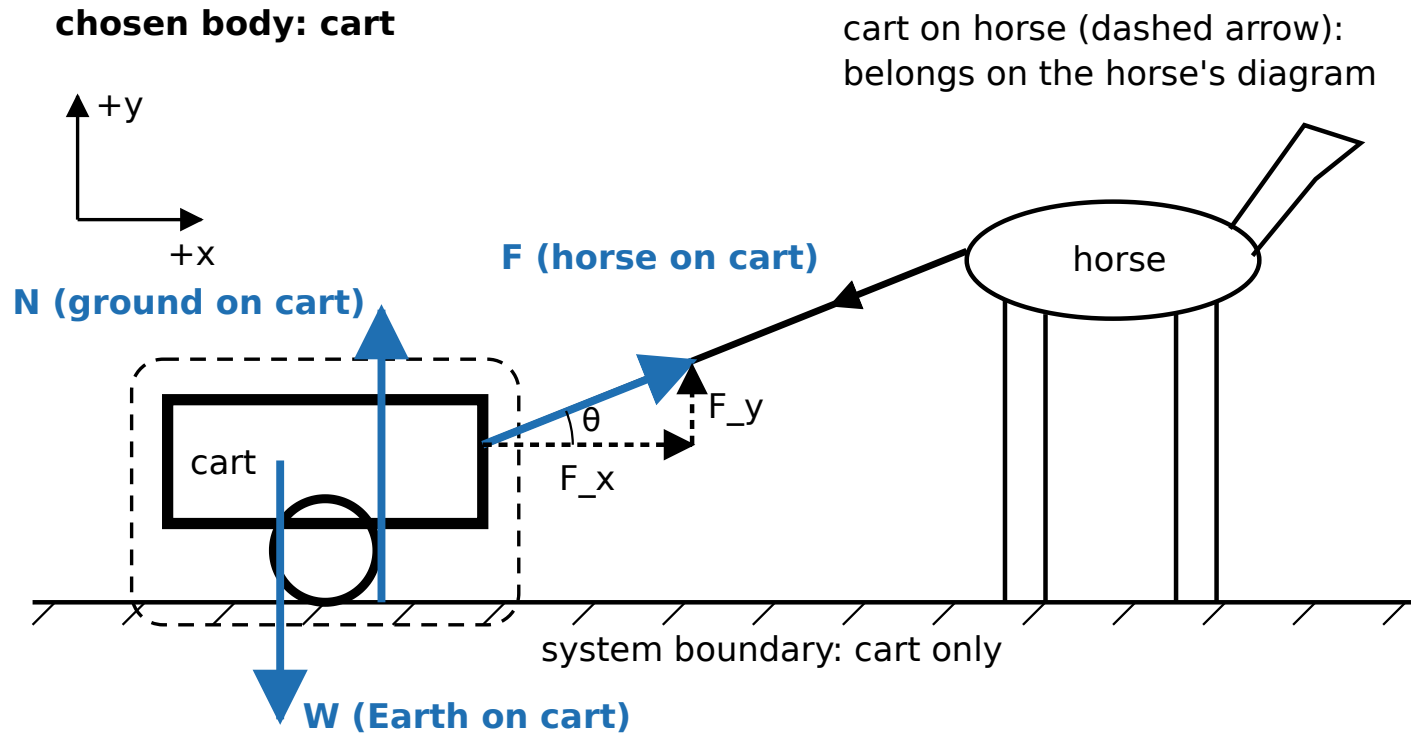
Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. State the connector model before using one symbol T . What does taut and inextensible imply about the two cart accelerations?

Reasoning route and full solution



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. The taut inextensible connector keeps the cart separation fixed, so their acceleration components along the track are equal.
2. Treating A+B as one system gives $F = (m_A + m_B)a$.
3. For B alone, the only horizontal force is the string tension, so $T = m_B a$.
4. Substitution gives $T = m_B F / (m_A + m_B)$.
5. Using one tension magnitude is a model consequence, not a universal property of every rope or pulley arrangement.

Answer

The string constraint gives one acceleration $a = F / (m_A + m_B)$. For B, $T = m_B a$, so $T = m_B F / (m_A + m_B)$. Equal tension relies on the ideal massless-string model; a connector with significant mass or a nonideal redirection can invalidate one common T.

Independent check

The tension is smaller than F for positive m_A and m_B , because part of F accelerates cart A as well as cart B.

If you went wrong

Your tension equals F : the string has only to accelerate cart B, so $T = m_B a$, which is less than F .

Repair

Revisit: Determine the tension magnitude from a chosen body's signed Newton-II equation; do not replace T by mg or by an applied force merely because those forces appear in the same setup.

What this establishes

Established: an ideal taut string gives one acceleration and one tension, and the tension comes from the body it pulls. The transfer task breaks one ideal assumption, a rough pulley axle, and asks what survives.

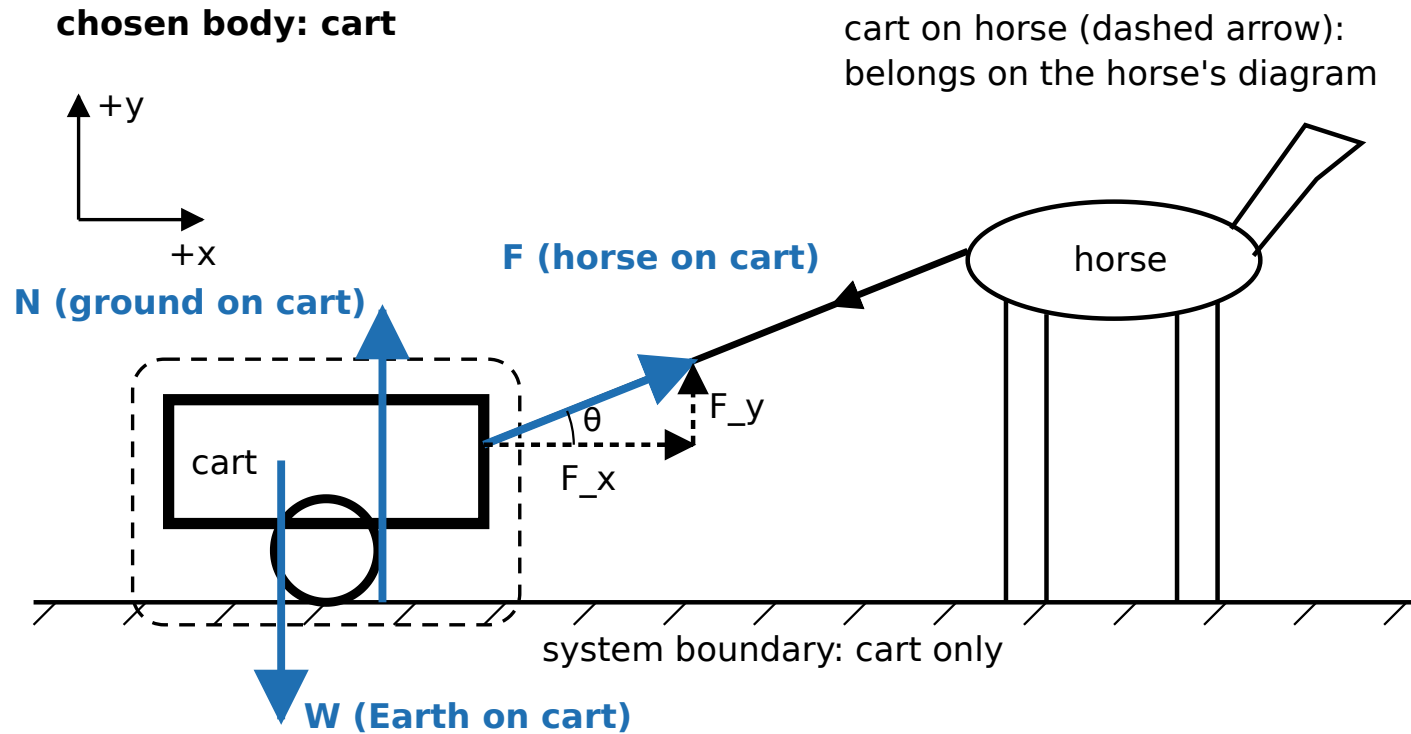
AUTHORED practice

Newton-law and observer-frame practice

One taut inextensible string passes over one fixed pulley. Let y_A and y_B be the two straight segment lengths measured positive away from the pulley. Write the string-length relation and derive the corresponding velocity and acceleration relations. If A accelerates away from the pulley with magnitude a , state B's signed acceleration in these coordinates.

Conditions

- The pulley is fixed.
- Only the two declared straight string segments change length.
- The string remains taut and inextensible.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

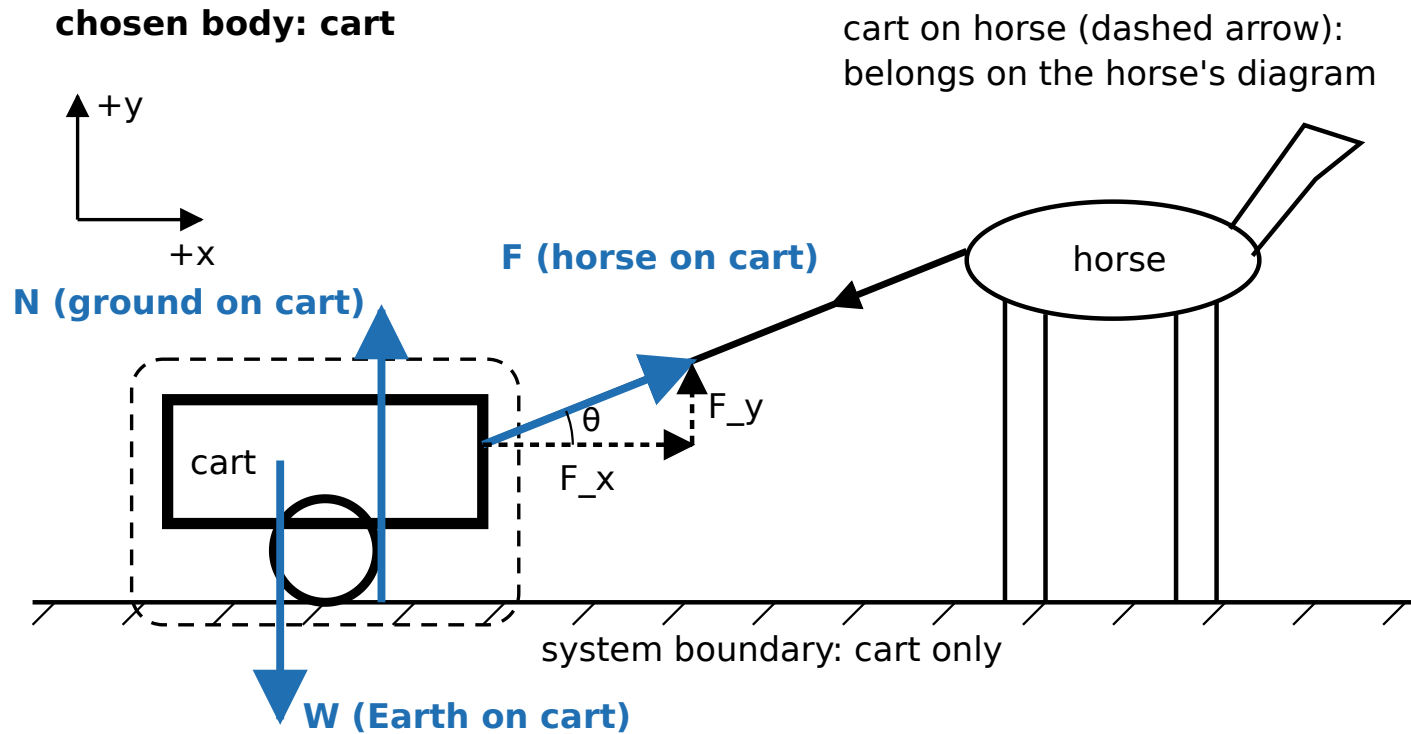
Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. Which parts of the string can change length while the pulley itself stays fixed?

Reasoning route and full solution



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. For one fixed pulley, the wrapped/fixed part of the string contributes no changing length in this bounded geometry.
2. Inextensibility makes the sum of the two changing straight segment lengths constant.
3. Taking one and two time derivatives gives the velocity and acceleration constraints.
4. With both coordinates positive away from the pulley, the signed accelerations are equal in magnitude and opposite in sign.

Answer

The variable string length obeys $y_A + y_B = \text{constant}$. Therefore $v_A + v_B = 0$ and $a_A + a_B = 0$. If $a_A = +a$ away from the pulley, then $a_B = -a$, toward the pulley.

Independent check

If both y_A and y_B increased together, total string length would increase, contradicting inextensibility.

If you went wrong

You wrote $a_A = a_B$: measured away from the pulley, one segment lengthens while the other shortens, so the signed accelerations must be opposite.

Repair

[Revisit: Differentiate the fixed-length relation at the kinematic level needed by the problem.](#)

What this establishes

Established: one fixed pulley and one inextensible string give $y_A + y_B = \text{constant}$, so $a_A + a_B = 0$ in away-from-pulley coordinates. The transfer task gives the geometry in words with upward coordinates, so the length relation must be rebuilt.

AUTHORED practice

Newton-law and observer-frame practice

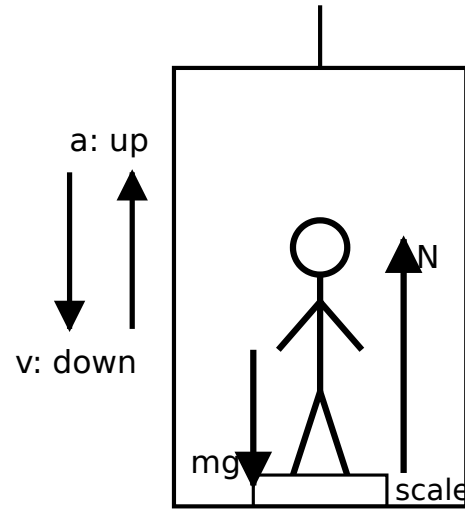
A bus accelerates to the right on a straight road. A loose object on a smooth horizontal shelf is described first by a roadside observer and then by a passenger in the bus. Explain how the two descriptions differ and where, if anywhere, a pseudo-force is used.

Conditions

- The roadside frame is treated as inertial.
- The bus has nonzero rightward acceleration.
- The object-shelf friction is negligible.

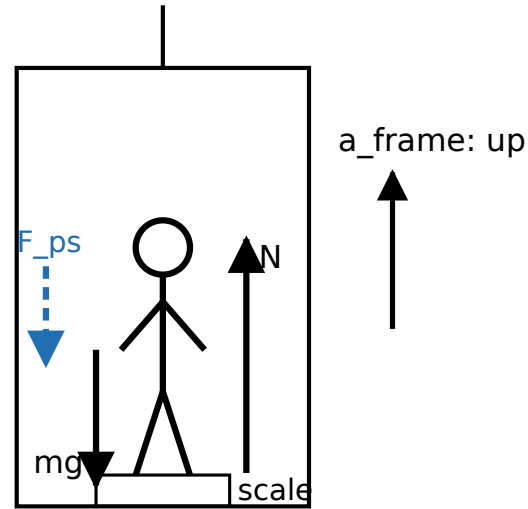
Ground observer (inertial)

$a_{\text{frame}} = 0$



Lift observer (accelerating frame)

person: $v = 0$, $a = 0$ relative to lift



chosen body: the person (same event in both columns)

Real (both): N (scale on person), up;
 mg (Earth on person), down.

$$N - mg = m a_y$$

contact: $N > 0$ (on the scale)

Dashed: $F_{ps} = -m a_{\text{frame}}$, down;
a frame term, no agent, no partner.

$$N - mg - m a_{\text{frame}} = 0$$

contact: $N > 0$ (on the scale)

Same observable in both frames: scale reads $N = m(g + a_y) > mg$.

Set by the upward acceleration, not by the downward velocity.

Stage 1 of 2

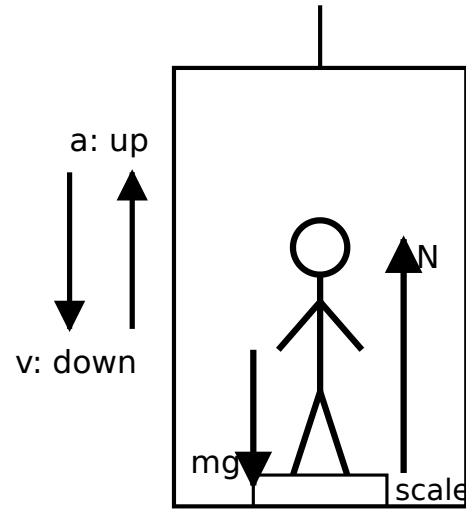
Show one physical event in an inertial and an accelerating observer description while keeping physical interactions fixed, making the pseudo-force visibly frame-dependent, and preserving the same observable scale reading or string direction.

1. Ask first whether any new physical object began touching or attracting the loose object when the bus started accelerating.

Reasoning route and full solution

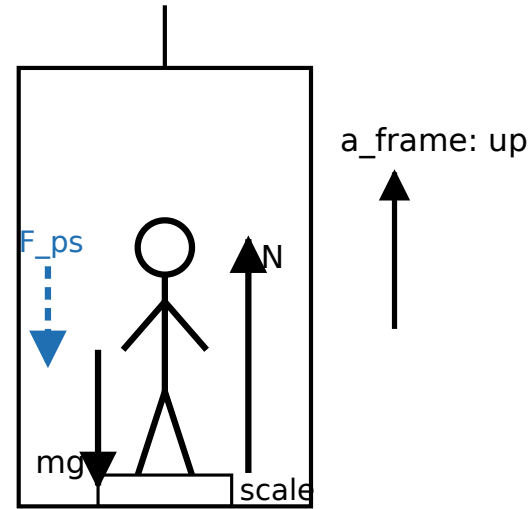
Ground observer (inertial)

$a_{\text{frame}} = 0$



Lift observer (accelerating frame)

person: $v = 0$, $a = 0$ relative to lift



chosen body: the person (same event in both columns)

Real (both): N (scale on person), up;
 mg (Earth on person), down.

$$N - mg = m a_y$$

contact: $N > 0$ (on the scale)

Dashed: $F_{ps} = -m a_{\text{frame}}$, down;
a frame term, no agent, no partner.

$$N - mg - m a_{\text{frame}} = 0$$

contact: $N > 0$ (on the scale)

Same observable in both frames: scale reads $N = m(g + a_y) > mg$.

Set by the upward acceleration, not by the downward velocity.

Stage 1 of 5

Show one physical event in an inertial and an accelerating observer description while keeping physical interactions fixed, making the pseudo-force visibly frame-dependent, and preserving the same observable scale reading or string direction.

1. Changing observer does not create a new physical agent acting on the loose object.
2. The roadside observer can describe the object with the physical forces only.
3. The passenger's frame accelerates with the bus, so relative to that observer the object appears to shift backward.
4. A backward pseudo-force is added only in the accelerating-frame description if that convention is used.

Answer

The roadside description uses only physical interactions; the loose object does not acquire a new backward physical force merely because the bus accelerates. In the accelerating bus frame, a backward pseudo-force may be introduced as a modelling convention so Newton's laws can be written in that non-inertial frame.

Independent check

If the bus later moves at constant velocity, changing from roadside to bus observer does not by itself require a pseudo-force.

If you went wrong

You gave the object a backward physical force in the roadside account: nothing new touches it; the backward term exists only in the passenger's accelerating description.

What this establishes

Established: an inertial observer uses physical forces only, and an accelerating observer adds $-m a_{\text{frame}}$. The transfer task asks you to choose the frame yourself from what the description must achieve.

AUTHORED practice

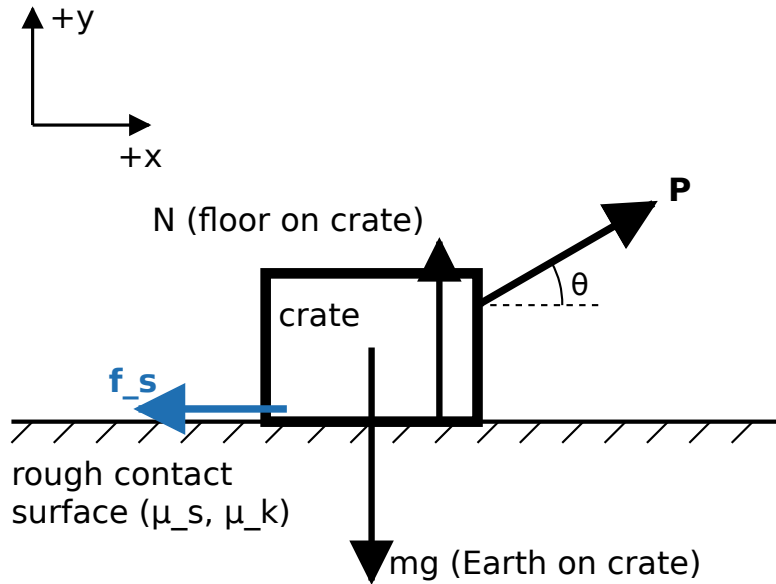
Newton-law and observer-frame practice

A crate rests on a rough horizontal floor. No horizontal force acts on it, and there is no tendency for the crate to slide relative to the floor. What is the friction force on the crate?

Conditions

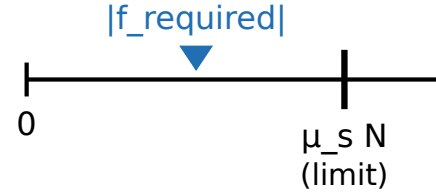
- The crate remains at rest.
- No other horizontal interaction acts on the crate.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



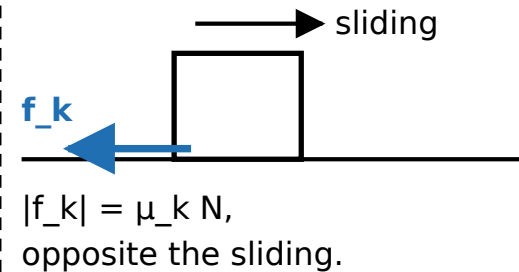
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$: Contact state: sliding



Stage 1 of 2

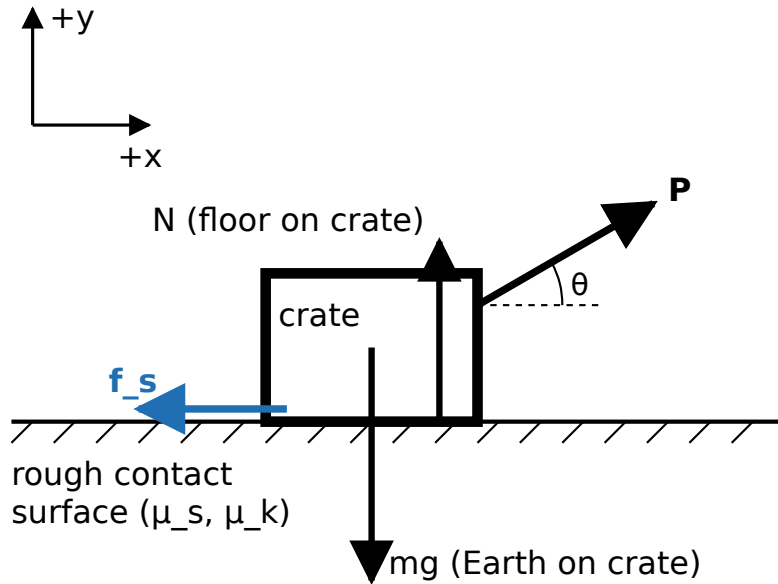
Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch to the

kinetic model only after sliding.

1. Ask what relative sliding or slip tendency would appear at the contact if friction were absent.

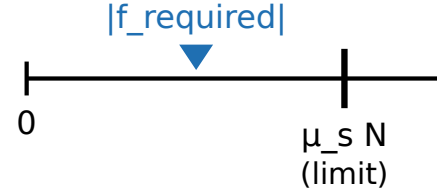
Reasoning route and full solution

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



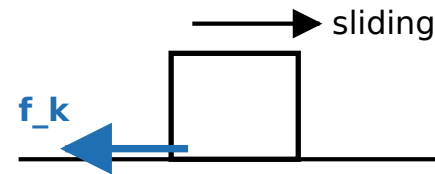
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Stage 1 of 5

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch

to the kinetic model only after sliding.

1. Friction responds to relative sliding or a tendency to slide at the contact.
2. Here no horizontal force or other effect creates a tangential slip tendency.
3. Therefore no tangential contact force is required, so $f=0$.

Answer

The friction force is zero.

Independent check

Roughness provides the ability to supply friction when needed; it does not require a nonzero friction force in every contact state.

If you went wrong

You gave a non-zero friction such as $\mu_s mg$: nothing pushes the crate sideways, so there is no slip tendency and nothing for friction to oppose.

Repair

[Revisit: Decide whether any tangential friction is required at all; allow friction to be zero when the contact has neither relative sliding nor a tendency to slide.](#)

What this establishes

Established: a rough contact supplies friction only when a slip tendency needs opposing, so a crate at rest with no sideways force has $f = 0$. The transfer task keeps this rule for a crate that moves together with its platform.

AUTHORED practice

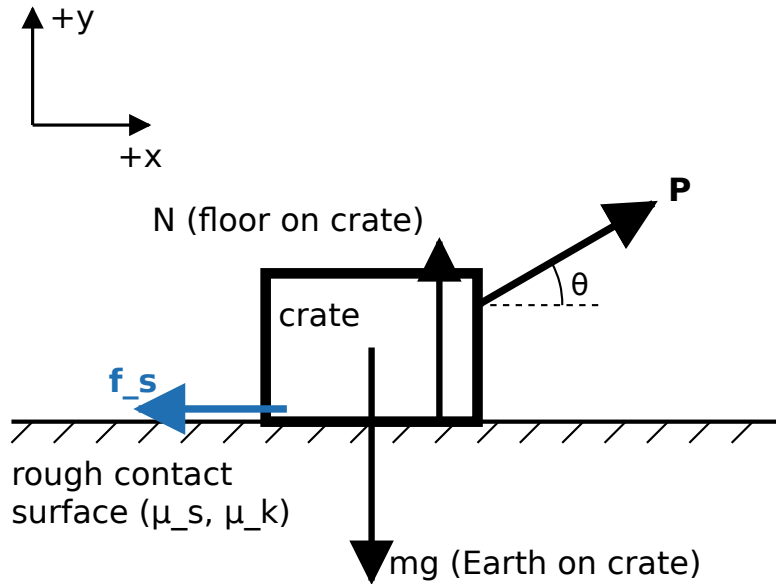
Newton-law and observer-frame practice

A 5 kg block rests on a rough horizontal floor. A 12 N horizontal pull is applied, but the block remains at rest. The coefficient of static friction is 0.6 and take $g=10 \text{ m/s}^2$. Find the actual static friction and its limiting value.

Conditions

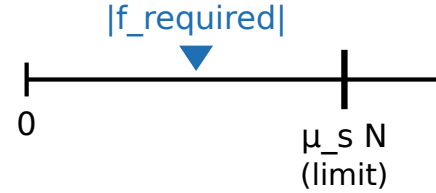
- The pull is horizontal.
- The block remains at rest.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



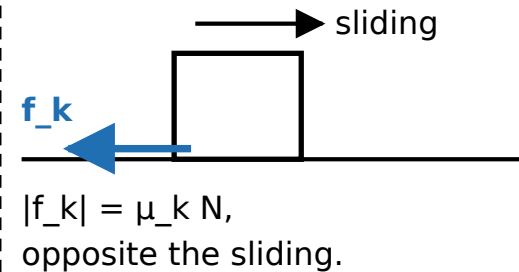
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Stage 1 of 2

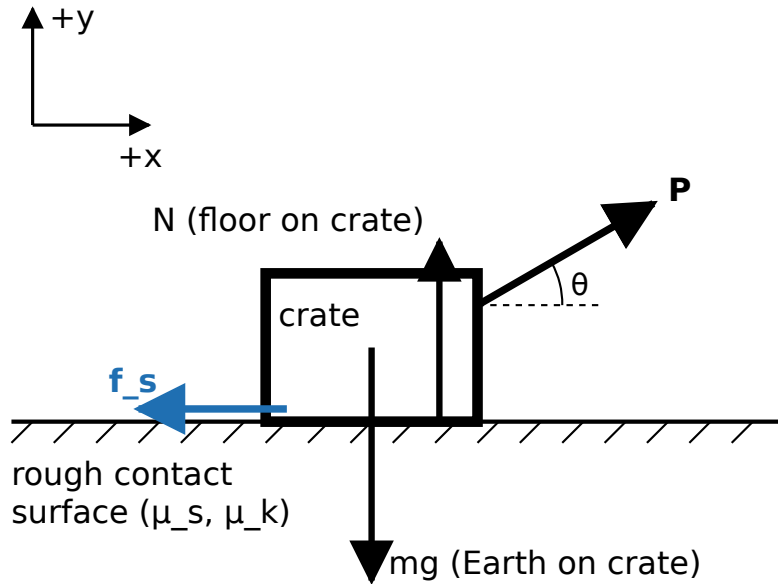
Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch to the

kinetic model only after sliding.

1. Because the block remains at rest, what horizontal force must friction balance?

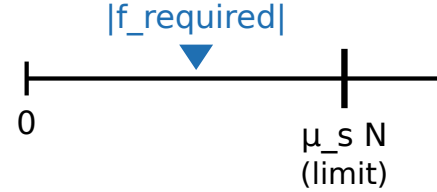
Reasoning route and full solution

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



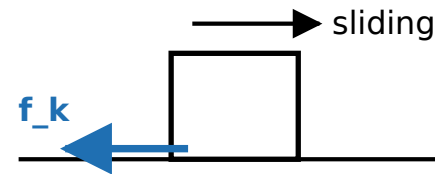
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Stage 1 of 5

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch

to the kinetic model only after sliding.

1. Because the block remains at rest, the horizontal net force is zero.
2. Static friction therefore supplies the 12 N needed to balance the pull, rather than automatically taking its maximum value.
3. The vertical forces give $N=mg=50$ N.
4. The limiting static friction is $\mu_s N=0.6(50)=30$ N, and 12 N is within that limit.

Answer

The actual static friction is 12 N opposite the pull; the limiting value is 30 N.

Independent check

The actual friction must change if the applied pull changes while the block remains static; the 30 N value is only the maximum available.

If you went wrong

Your friction is 30 N: that is the most static friction can give, but a block at rest under a 12 N pull needs only 12 N; 30 N would drive it backward.

Repair

Revisit: Assume no relative sliding provisionally and solve the tangential Newton-II equation for the static friction value f_{required} that would make that motion possible.

What this establishes

Established: static friction takes whatever value rest requires, up to $\mu_s N$; the limit is a ceiling, not the value. The transfer task tilts the push, so the ceiling itself changes with the force.

AUTHORED practice

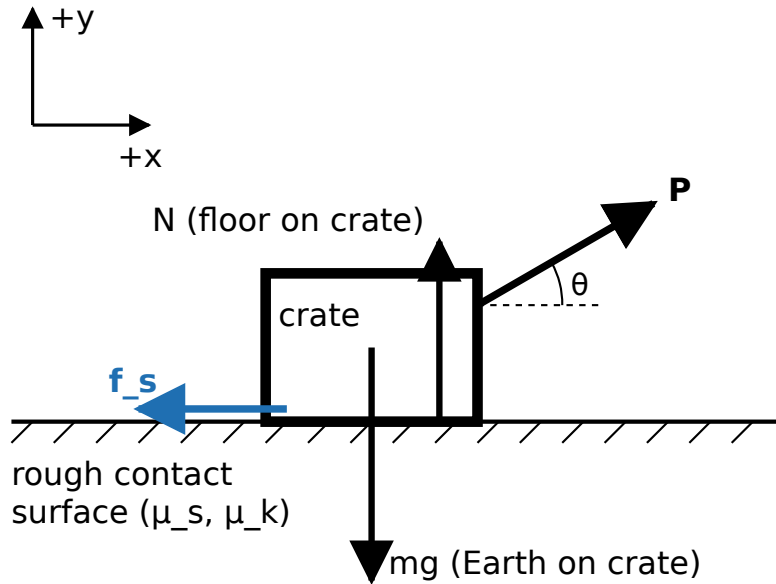
Newton-law and observer-frame practice

A 5 kg block is already sliding to the right on a horizontal floor. A constant 18 N horizontal force pulls it to the right. The coefficient of kinetic friction is 0.20 and take $g=10 \text{ m/s}^2$. Find the acceleration.

Conditions

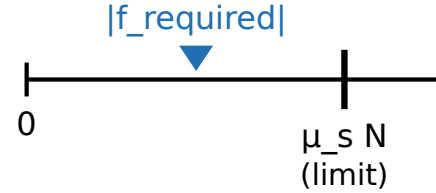
- Relative sliding is already established.
- The applied force is horizontal.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



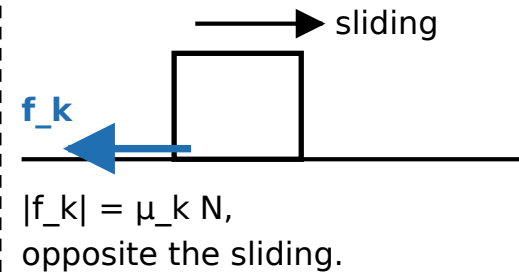
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



Stage 1 of 2

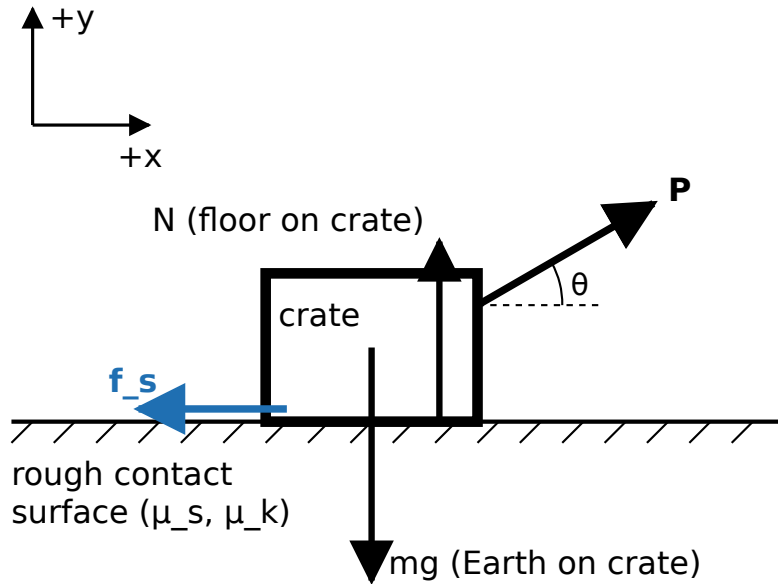
Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch to the

kinetic model only after sliding.

1. The contact state is already given: which friction model belongs to established sliding?

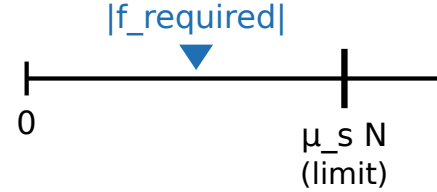
Reasoning route and full solution

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



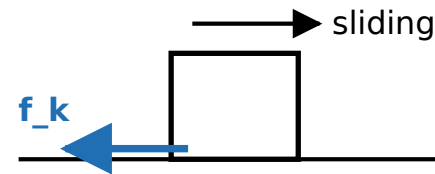
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Stage 1 of 5

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch

to the kinetic model only after sliding.

1. The vertical force balance gives $N=mg=50\text{ N}$.
2. Because the block is sliding, kinetic friction has magnitude $\mu_k N=0.20(50)=10\text{ N}$ and points left, opposite the relative sliding.
3. The horizontal net force is $18-10=8\text{ N}$ to the right.
4. Newton's second law gives $a=8/5=1.6\text{ m/s}^2$ to the right.

Answer

The block accelerates at 1.6 m/s^2 to the right.

Independent check

The kinetic model is justified by the stated sliding; no static-friction feasibility test is needed for this already-sliding state.

If you went wrong

Your acceleration is 3.6 m/s^2 ($18/5$, friction left out) or 5.6 m/s^2 (friction added to the pull): kinetic friction of 10 N acts against the sliding and must be subtracted.

Repair

Revisit: Use $|f_s| = \mu_s N$ only at impending slip; if relative sliding is already established, use $|f_k| = \mu_k N$ with direction opposing the relative sliding.

What this establishes

Established: once sliding is given, friction is fixed at $\mu_k N$ opposite the sliding, and Newton II gives the acceleration. The transfer task moves the applied force onto one block of a stacked pair, so the no-slip test has to be rebuilt.

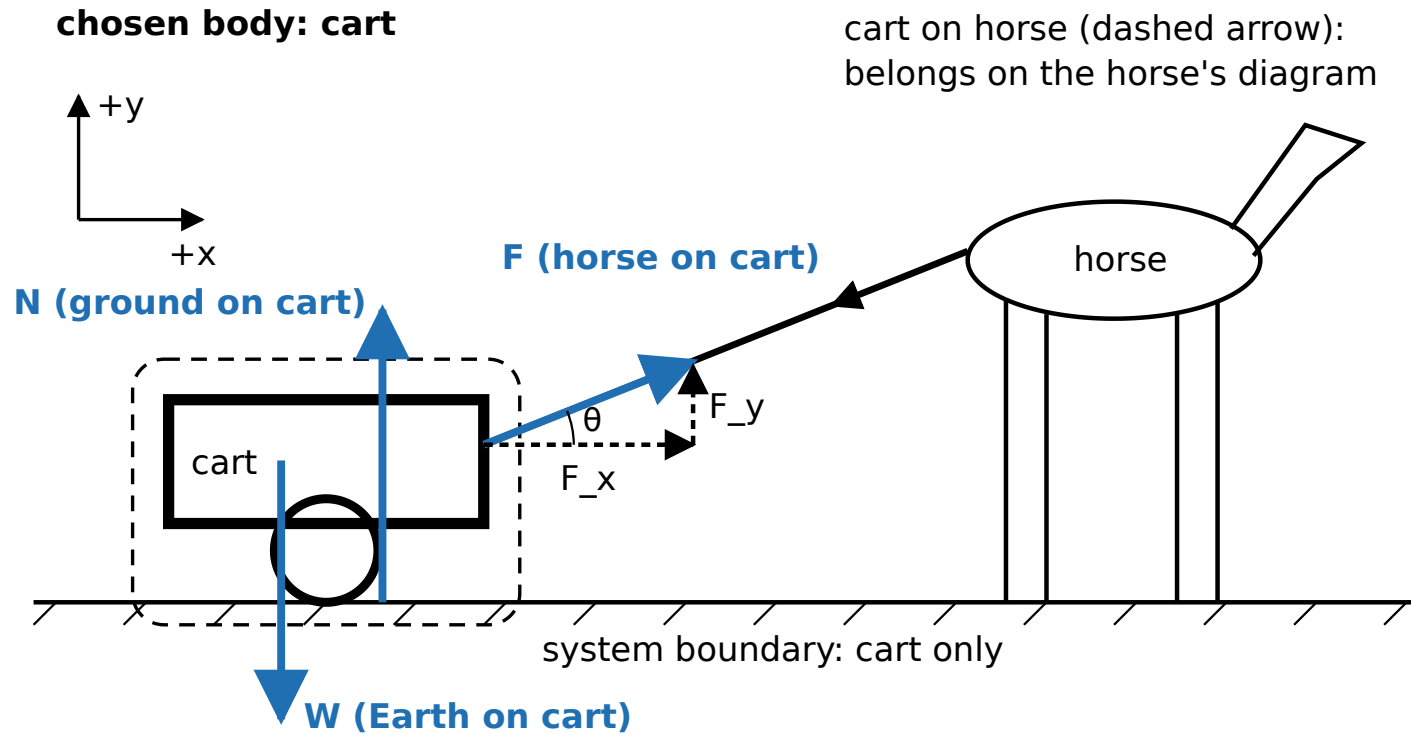
AUTHORED practice

Newton-law and observer-frame practice

Blocks A and B are side by side on a smooth horizontal floor, with A to the left of B and no adhesive connection. Compare two cases. In case 1 a horizontal force pushes A to the right into B and contact is maintained. In case 2 the same force pulls A to the left, away from B. In which case may one common horizontal acceleration be imposed, and why?

Conditions

- The contact force is an ordinary normal force and cannot pull the blocks together.
- The floor is frictionless.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

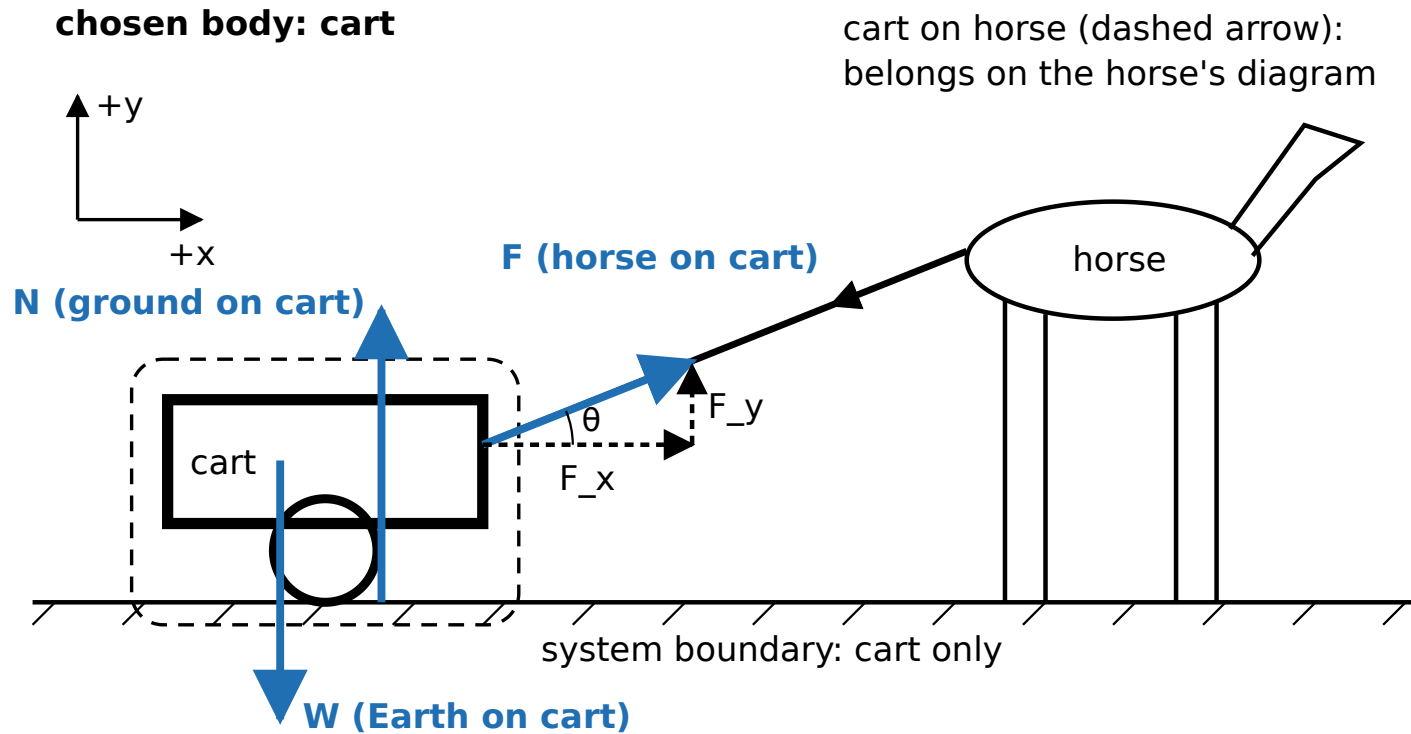
Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. What physical condition must stay active before equal acceleration can be imposed?

Reasoning route and full solution



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. Common acceleration must be earned from an active constraint that fixes relative separation.
2. When A is pushed into B and contact is maintained, the normal interaction can enforce the shared motion.
3. When A is pulled away from B, an ordinary normal force cannot act as a tensile connector to keep them together.
4. After separation, each block's acceleration must be obtained from its own remaining forces.

Answer

A common acceleration is justified in case 1 while maintained contact fixes the separation. It is not justified in case 2: the normal contact cannot pull B after A is pulled away, so the contact constraint is lost.

Independent check

If a proposed case-2 solution assigns a nonzero pulling normal force on B to preserve contact, it has given an ordinary contact force a capability it does not have.

If you went wrong

You gave both cases the acceleration $F/(m_A + m_B)$: in case 2 that would need an ordinary contact to pull B along, and a contact can only push.

Repair

[Revisit: Identify the connection/contact condition and state exactly which relative separation is fixed.](#)

What this establishes

Established: a common acceleration holds only while a real constraint keeps the separation fixed, and a pushing contact cannot pull. The transfer task asks what each block does once the contact is actually lost.

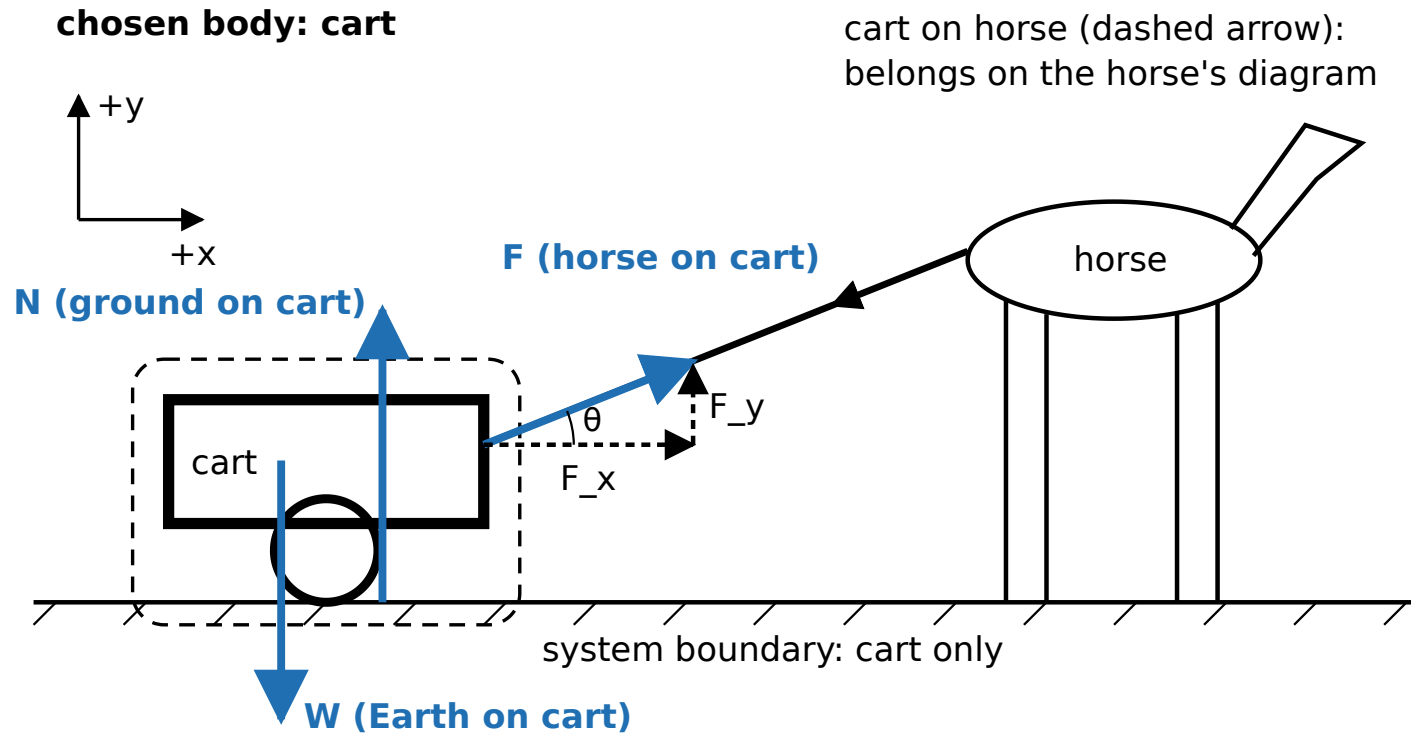
AUTHORED practice

Newton-law and observer-frame practice

Masses m_A and m_B , with $m_B > m_A$, hang from the two ends of one taut massless inextensible string over one fixed frictionless massless pulley. Derive the acceleration magnitude a and the common string tension T . State explicitly why T is not equal to either hanging weight during the acceleration.

Conditions

- Mass A accelerates upward and mass B accelerates downward.
- Use the same acceleration magnitude a for the two ends, as required by the fixed string-length constraint.
- Neglect air resistance.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

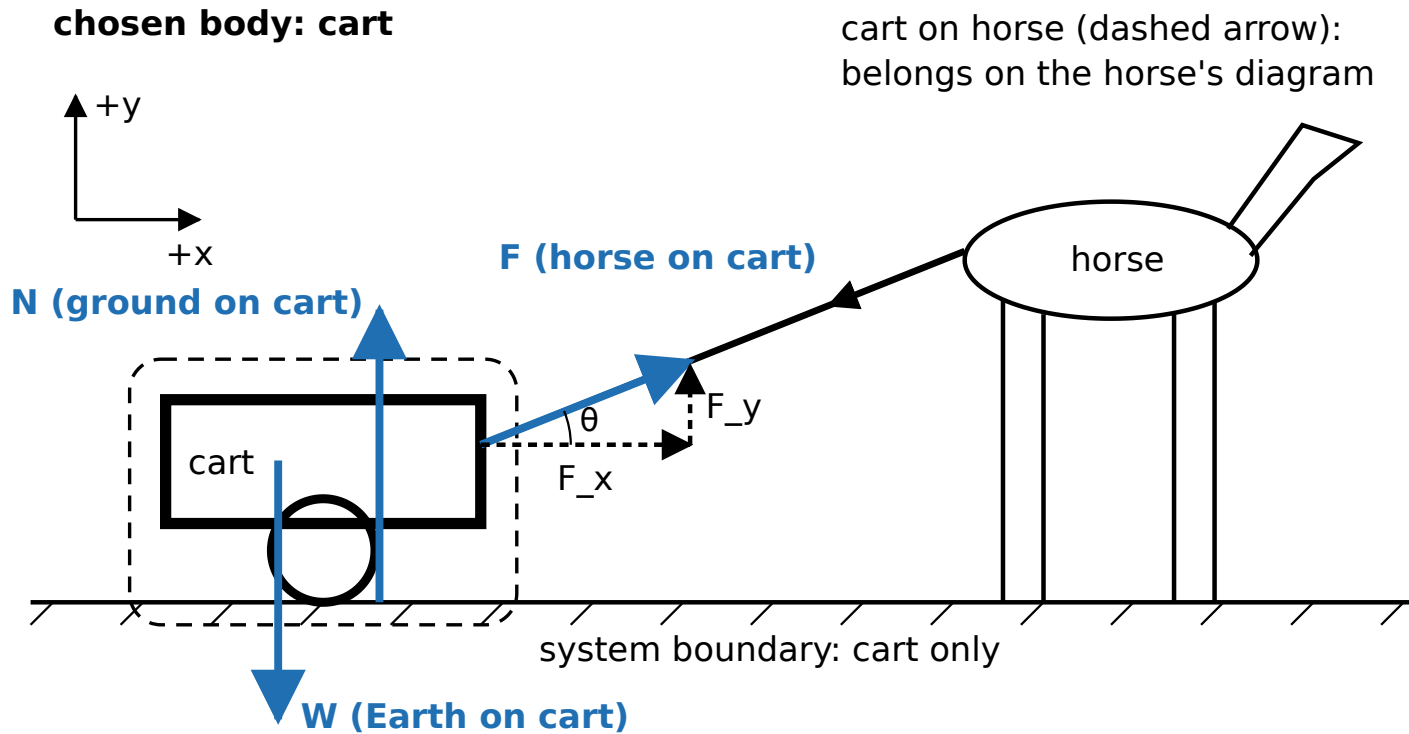
Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. Separate the two claims you need: what fixes the acceleration relationship, and what licenses one common tension magnitude?

Reasoning route and full solution



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. The fixed inextensible string gives equal acceleration magnitudes for the two ends while the ideal string/pulley model licenses one common tension T .
2. For A, upward positive gives $T - m_A g = m_A a$.
3. For B, downward positive gives $m_B g - T = m_B a$.
4. Adding the equations gives $a = (m_B - m_A)g / (m_A + m_B)$.
5. Substitution into either body equation gives $T = 2m_A m_B g / (m_A + m_B)$.
6. Because A accelerates upward, its tension exceeds $m_A g$; because B accelerates downward, its tension is below $m_B g$.

Answer

The acceleration magnitude is $a = (m_B - m_A)g / (m_A + m_B)$, and the common tension is $T = 2m_A m_B g / (m_A + m_B)$. During the motion $T > m_A g$ and $T < m_B g$, so it is not equal to either weight unless the corresponding acceleration is zero.

Independent check

If $m_A = m_B$, the formula reduces to $a = 0$ and $T = mg$, recovering the equilibrium special case.

If you went wrong

Your tension is $m_A g$ or $m_B g$: while the masses accelerate, the string pulls A harder than A's weight and pulls B less than B's weight.

Repair

Revisit: Determine the tension magnitude from a chosen body's signed Newton-II equation; do not replace T by mg or by an applied force merely because those forces appear in the same setup.

What this establishes

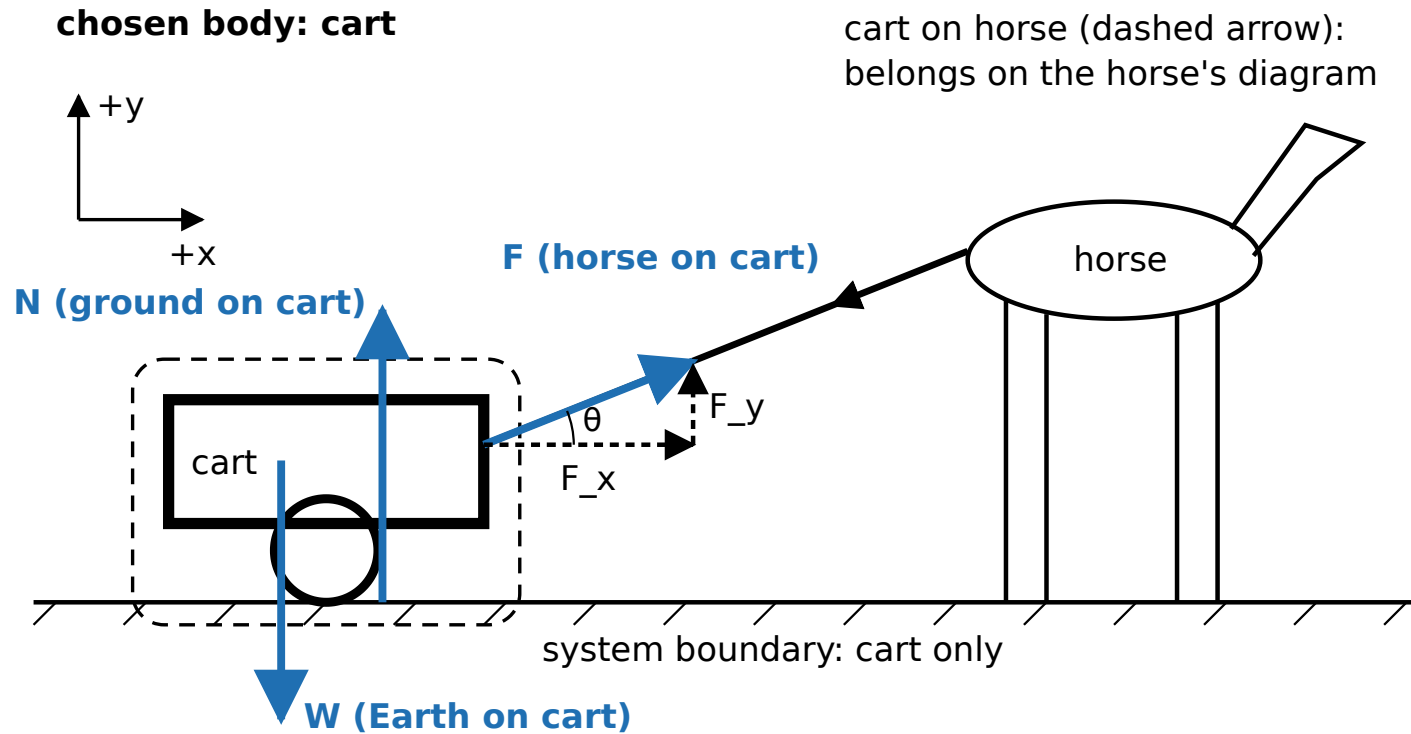
Established: the fixed string length gives equal acceleration magnitudes, the ideal pulley gives one T , and each mass's own equation places T between the two weights. The transfer task makes the pulley heavy, so these two claims have to be separated.

Inclined-plane force modelling

A block of mass m is released from rest on a smooth fixed plane inclined at angle θ above the horizontal. Choose axes that make the contact constraint simple, determine the normal reaction, and find the block's acceleration along the plane.

Conditions

- $0 < \theta < 90$ degrees.
- The plane is fixed and smooth.
- The block remains in contact with the plane.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

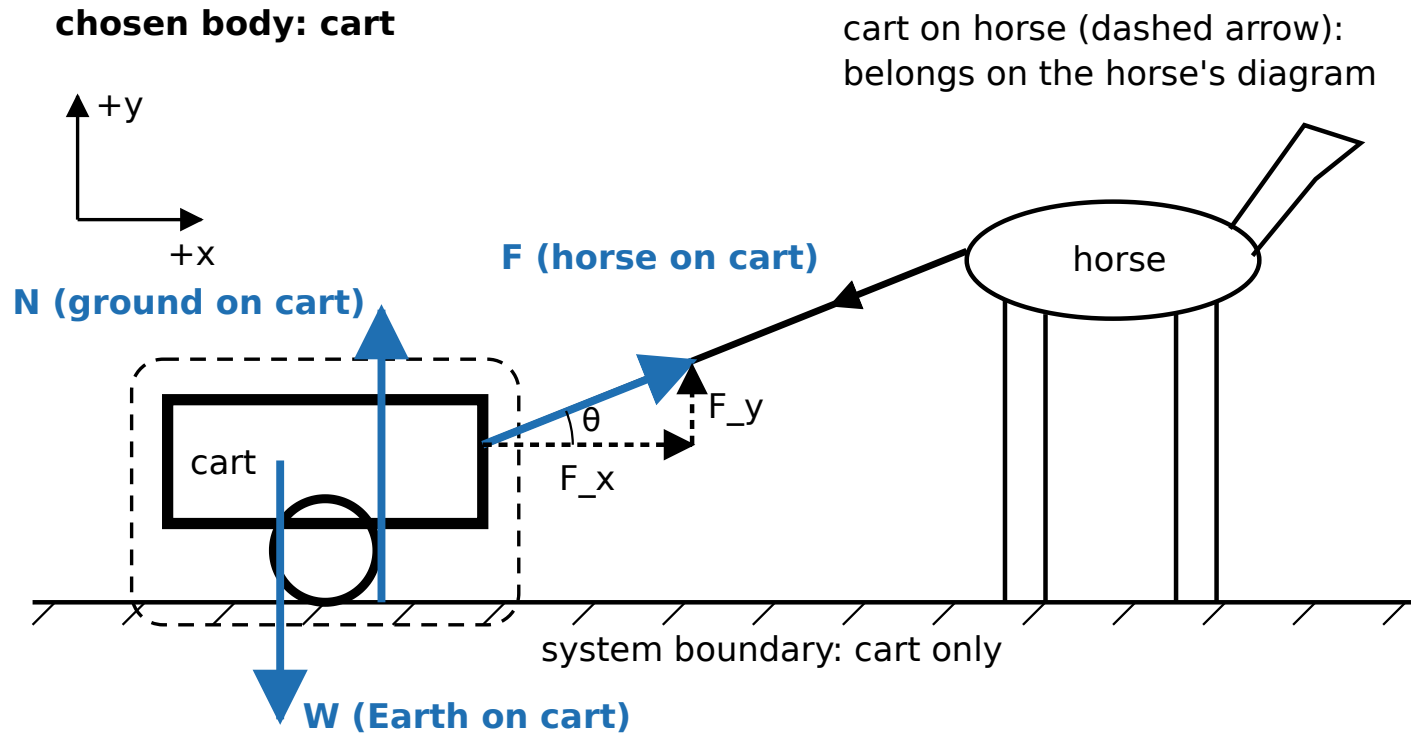
Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. Which axis choice turns maintained contact into one simple acceleration statement?

Reasoning route and full solution



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. Choose the block and use +parallel down the plane and +perpendicular outward from the plane.
2. The physical forces are weight mg and the normal reaction N .
3. Resolve weight into $mg \sin(\theta)$ parallel down the plane and $mg \cos(\theta)$ perpendicular into the plane.
4. Maintained contact gives $a_{\text{perp}}=0$, so $N-mg \cos(\theta)=0$.
5. Along the plane, $mg \sin(\theta)=m a_{\text{parallel}}$, so $a_{\text{parallel}}=g \sin(\theta)$ down the plane.

Answer

Using axes parallel/perpendicular to the plane, weight has components $mg \sin(\theta)$ down the plane and $mg \cos(\theta)$ into the plane. Maintained contact gives $N=mg \cos(\theta)$, and the tangential equation gives acceleration $g \sin(\theta)$ down the plane.

Independent check

At $\theta=0$ the tangential acceleration tends to zero and N tends to mg ; as θ approaches 90 degrees, N tends to zero and the tangential acceleration tends to g .

If you went wrong

You wrote $N = mg$ or $a = g$: the plane supports only the perpendicular part of the weight, $mg \cos \theta$, and only $mg \sin \theta$ drives the block along it.

Repair

Revisit: Choose axes that make the active constraint easy to express; for a fixed inclined contact, use parallel/perpendicular axes when helpful, resolve the physical forces into those axes, and write separate signed Newton-II equations.

What this establishes

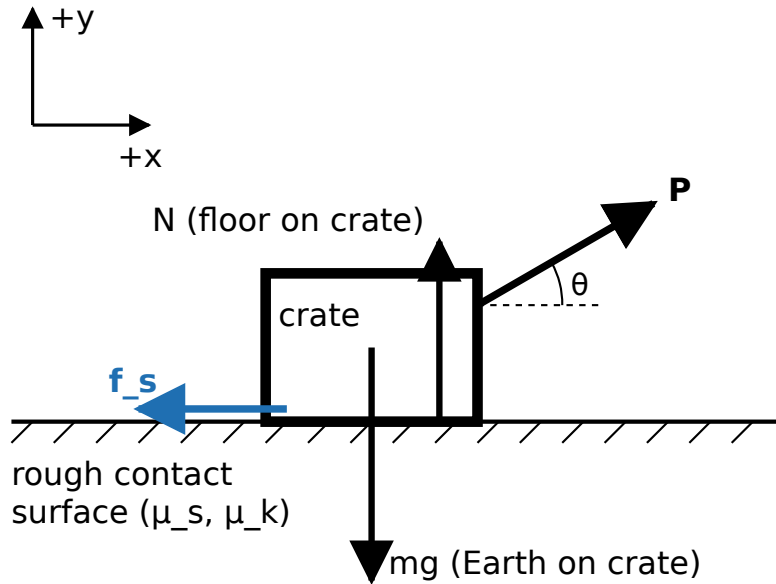
Established: contact-aligned axes turn maintained contact into $a_{\text{perp}} = 0$, giving $N = mg \cos \theta$ and $a = g \sin \theta$. The transfer task compares this with laboratory axes to show that both describe the same physics.

A block rests on a rough fixed plane inclined at angle θ . The coefficient of static friction is μ_s . Determine the friction required for rest and derive the condition under which the static state is possible.

Conditions

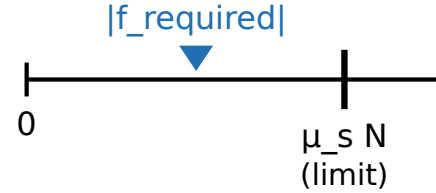
- No other force acts on the block.
- The block is tested for rest; do not assume limiting friction at the start.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



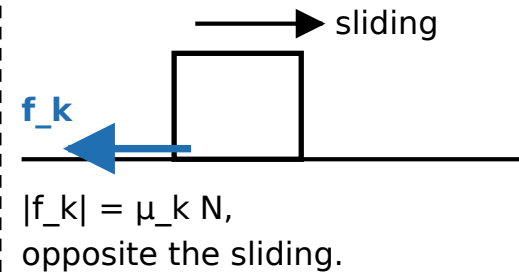
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



Stage 1 of 2

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch to the

kinetic model only after sliding.

1. First ask which way the block would tend to slip if friction vanished.

Reasoning route and full solution

1. **REPRESENT** Choose axes parallel and perpendicular to the plane, identify the block as the body, and determine the no-friction slip tendency.
Why valid: The contact-aligned frame makes the normal and tangential force requirements explicit, while the no-friction tendency determines which way static friction would have to act to maintain rest.
2. **TRANSFORM** Apply perpendicular equilibrium to obtain $N = mg \cos(\theta)$.
Why valid: For this item no other force has a perpendicular component and the block remains in contact without perpendicular acceleration.
3. **CONNECT** Assume rest provisionally and use tangential equilibrium to find the friction required by that state: $f_{\text{required}} = mg \sin(\theta)$.
Why valid: Static friction responds to the force demand of the no-slip state; its limiting value is a bound to test afterward, not the starting equality.
4. **DECIDE** Compare the required friction magnitude with the static limit and accept the rest model only if $|f_{\text{required}}| \leq \mu_s N$.

Why valid: The canonical static-friction relation permits the required no-slip value only while its magnitude stays within $\mu_s N$.

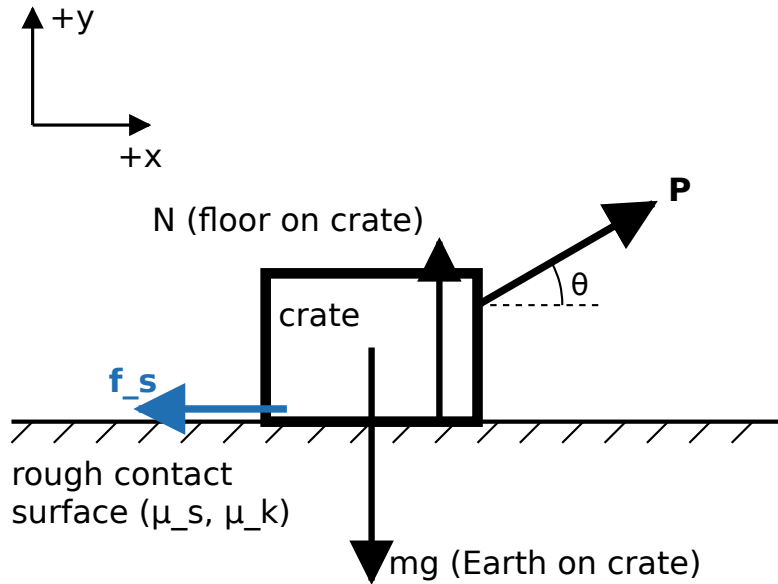
5. **TRANSFORM** Simplify the feasibility inequality to $\tan(\theta) \leq \mu_s$.

Why valid: For $0 \leq \theta < 90$ degrees, $mg \cos(\theta)$ is positive, so division preserves the inequality direction.

6. **VERIFY** Check that equality is used only at impending slip and that smaller angles require static friction below its maximum.

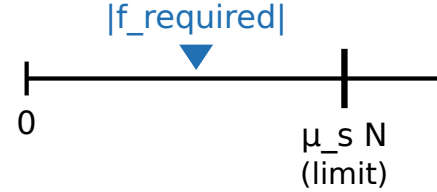
Why valid: A limiting equality everywhere would contradict the static-friction bound and the zero/small-demand cases.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



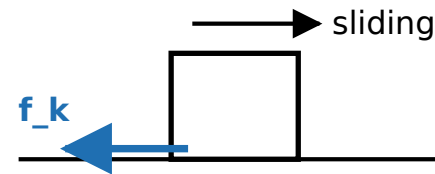
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Stage 1 of 5

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch

to the kinetic model only after sliding.

1. Without friction the block tends to slide down the plane, so static friction required for rest points up the plane.
2. The perpendicular equation gives $N = mg \cos(\theta)$.
3. The no-slip tangential equation requires $f_{\text{required}} = mg \sin(\theta)$.
4. Static contact is possible only if $f_{\text{required}} \leq \mu_s N$.
5. Cancelling $mg \cos(\theta) > 0$ gives $\tan(\theta) \leq \mu_s$.

Answer

For rest, $N = mg \cos(\theta)$ and static friction must act up the plane with required magnitude $mg \sin(\theta)$. Rest is feasible when $mg \sin(\theta) \leq \mu_s mg \cos(\theta)$, equivalently $\tan(\theta) \leq \mu_s$.

Independent check

The calculation uses equality $f_s = \mu_s N$ only at the limiting angle; for a smaller angle the actual static friction is below its maximum.

If you went wrong

You set the friction to $\mu_s mg \cos \theta$ at the start: that is only its limit. Rest needs $mg \sin \theta$, and that need is what gets compared with the limit.

Repair

[Revisit: Compare \$|f_{\text{required}}|\$ with \$\mu_s N\$.](#)

What this establishes

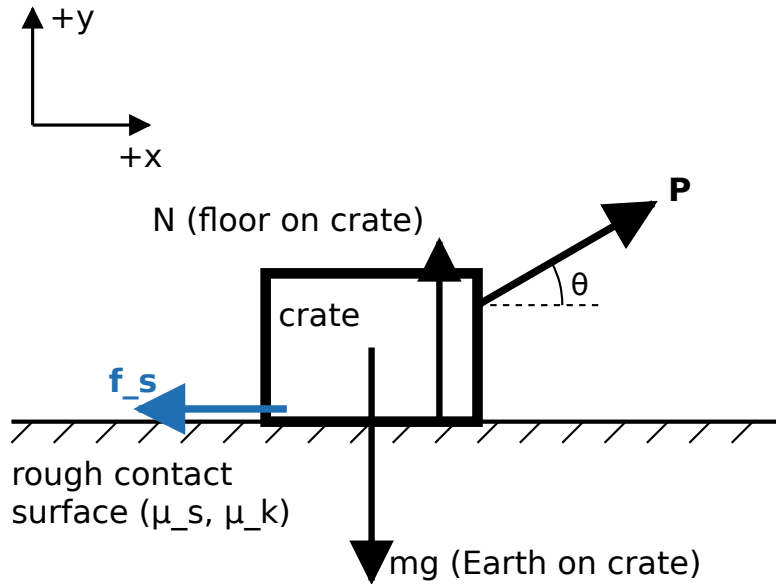
Established: rest on an incline needs friction $mg \sin \theta$ up the slope, which is possible only while $\tan \theta \leq \mu_s$. The transfer task adds a push along the slope, so the needed friction can shrink to zero or reverse.

A block is already sliding down a rough fixed plane inclined at angle θ . The coefficient of kinetic friction is μ_k . Derive its acceleration along the plane.

Conditions

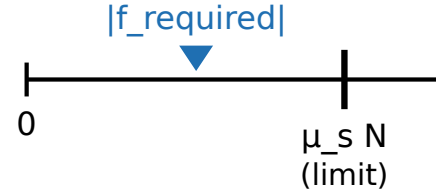
- The sliding direction is down the plane.
- No other force acts on the block.
- The plane is fixed.

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



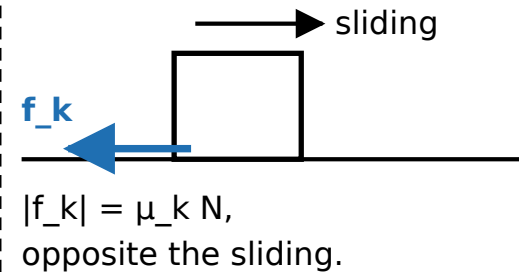
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



Stage 1 of 2

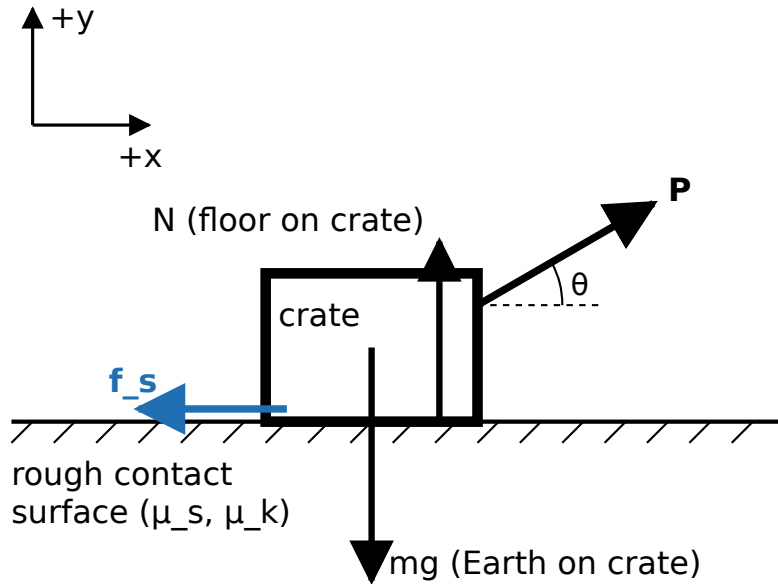
Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch to the

kinetic model only after sliding.

1. The block is already sliding down the plane; which way must the kinetic friction act at the contact?

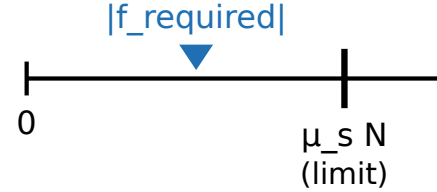
Reasoning route and full solution

chosen body: crate



N comes from the y equation;
it is not assumed equal to mg .

Friction demand vs limit



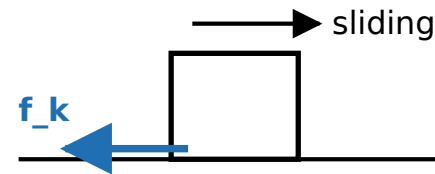
Contact state: static (no slip)

feasible while $|f_{\text{required}}| \leq \mu_s N$;

then $f_s = f_{\text{required}}$.

$|f_s| = \mu_s N$ holds only
at impending slip.

If $|f_{\text{required}}| > \mu_s N$:
Contact state: sliding



$|f_k| = \mu_k N$,
opposite the sliding.

Stage 1 of 5

Make static friction a responsive contact force rather than an automatic μN value: construct the chosen-body force account, determine the no-slip demand, compare it with the static limit, and switch

to the kinetic model only after sliding.

1. Maintained contact gives $N = mg \cos(\theta)$.
2. Because the block is sliding down relative to the plane, kinetic friction points up the plane.
3. With down-plane positive, $ma = mg \sin(\theta) - \mu_k mg \cos(\theta)$.
4. Divide by m to obtain $a = g[\sin(\theta) - \mu_k \cos(\theta)]$.

Answer

$N = mg \cos(\theta)$. Kinetic friction has magnitude $\mu_k mg \cos(\theta)$ and points up the plane, so the down-plane acceleration is $a = g[\sin(\theta) - \mu_k \cos(\theta)]$.

Independent check

Setting $\mu_k = 0$ recovers the smooth-plane result $a = g \sin(\theta)$.

If you went wrong

Your acceleration is $g(\sin \theta + \mu_k \cos \theta)$: friction was added along the sliding, which would make the rough slope faster than a smooth one.

Repair

Revisit: Use $|f_s| = \mu_s N$ only at impending slip; if relative sliding is already established, use $|f_k| = \mu_k N$ with direction opposing the relative sliding.

What this establishes

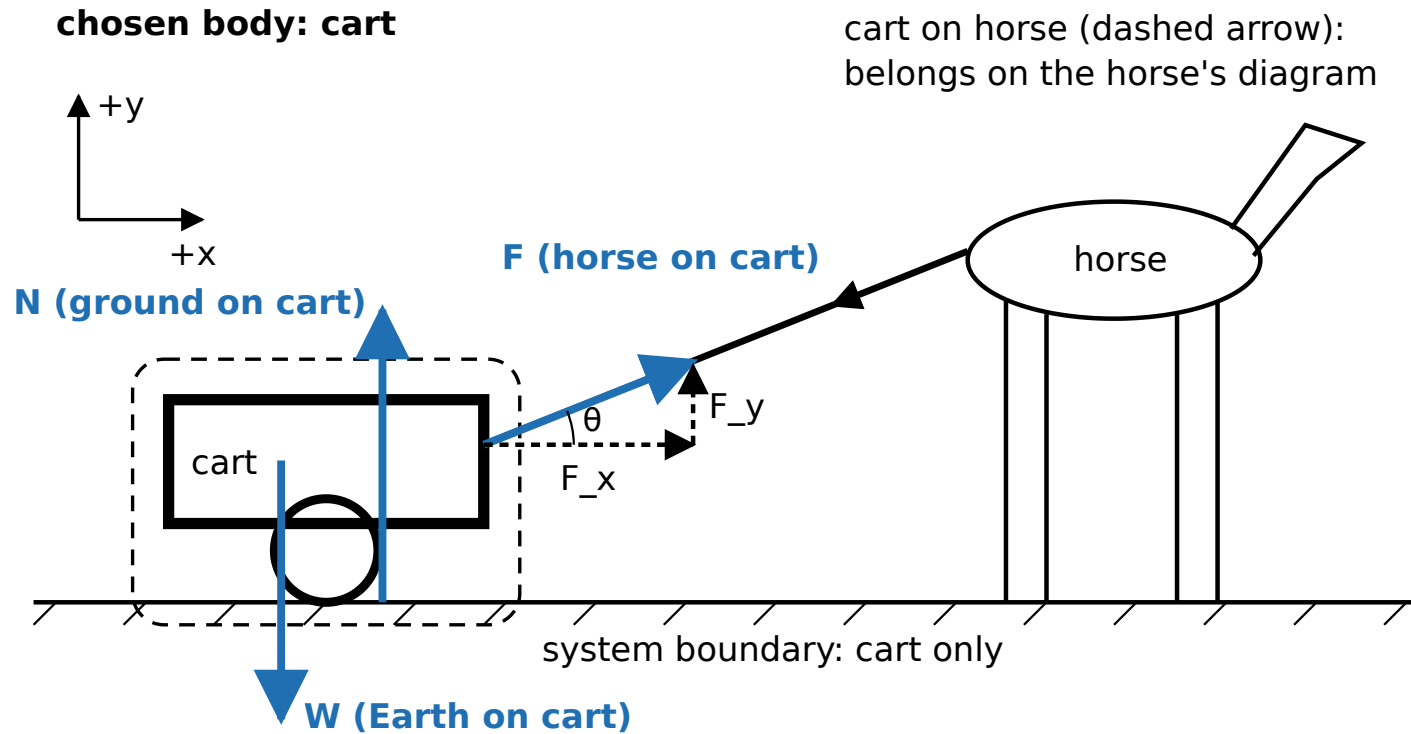
Established: on a fixed incline $N = mg \cos \theta$, and kinetic friction up the slope reduces the acceleration to $g(\sin \theta - \mu_k \cos \theta)$. The transfer task uses this acceleration inside a timing comparison to infer μ_k .

Inclined-plane force modelling

A block of mass m is on a smooth fixed plane that rises to the right at angle θ above the horizontal. A horizontal force P pushes the block to the right. Determine the normal reaction and the acceleration along the plane.

Conditions

- $0 < \theta < 90$ degrees.
- The block remains in contact with the smooth plane.
- The applied force P is horizontal and rightward.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

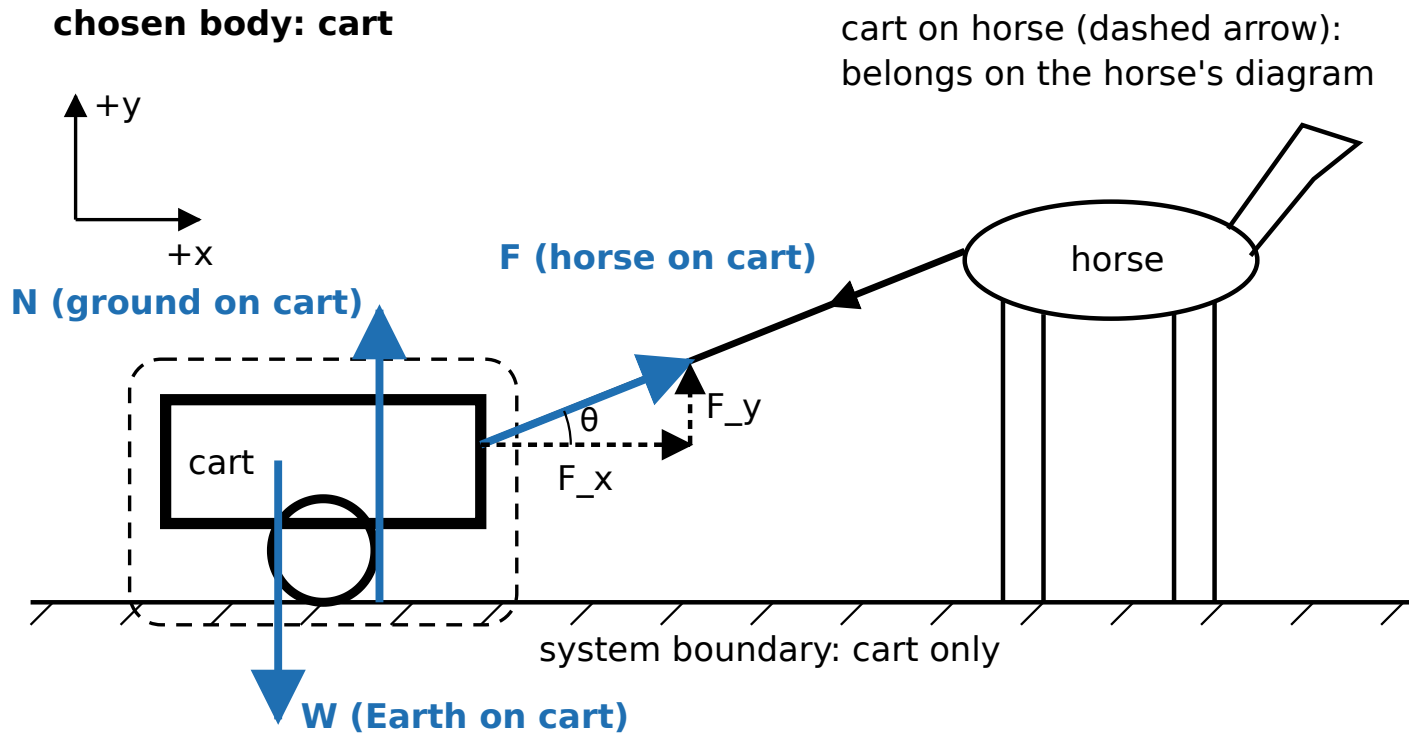
Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. The horizontal applied force is not parallel to the plane. Which component presses into the plane and which acts along it?

Reasoning route and full solution



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. Choose +parallel uphill and +perpendicular outward from the plane.
2. Resolve the horizontal force into $P \cos(\theta)$ uphill and $P \sin(\theta)$ into the plane.
3. Resolve weight into $mg \sin(\theta)$ downhill and $mg \cos(\theta)$ into the plane.
4. Maintained contact gives $N - mg \cos(\theta) - P \sin(\theta) = 0$.
5. The parallel equation is $P \cos(\theta) - mg \sin(\theta) = m a_{\text{parallel}}$.

Answer

Relative to plane-aligned axes, P has $P \cos(\theta)$ up the plane and $P \sin(\theta)$ into the plane. Thus $N = mg \cos(\theta) + P \sin(\theta)$, and $a_{\text{parallel}} = [P \cos(\theta) - mg \sin(\theta)]/m$ with uphill positive.

Independent check

At $P=0$ the result reduces to the ordinary smooth-incline equations, with negative uphill acceleration $-g \sin(\theta)$.

If you went wrong

You wrote $N = mg \cos \theta$: the horizontal push also presses the block into the plane and adds $P \sin \theta$ to the normal force.

Repair

Revisit: Choose axes that make the active constraint easy to express; for a fixed inclined contact, use parallel/perpendicular axes when helpful, resolve the physical forces into those axes, and write separate signed Newton-II equations.

What this establishes

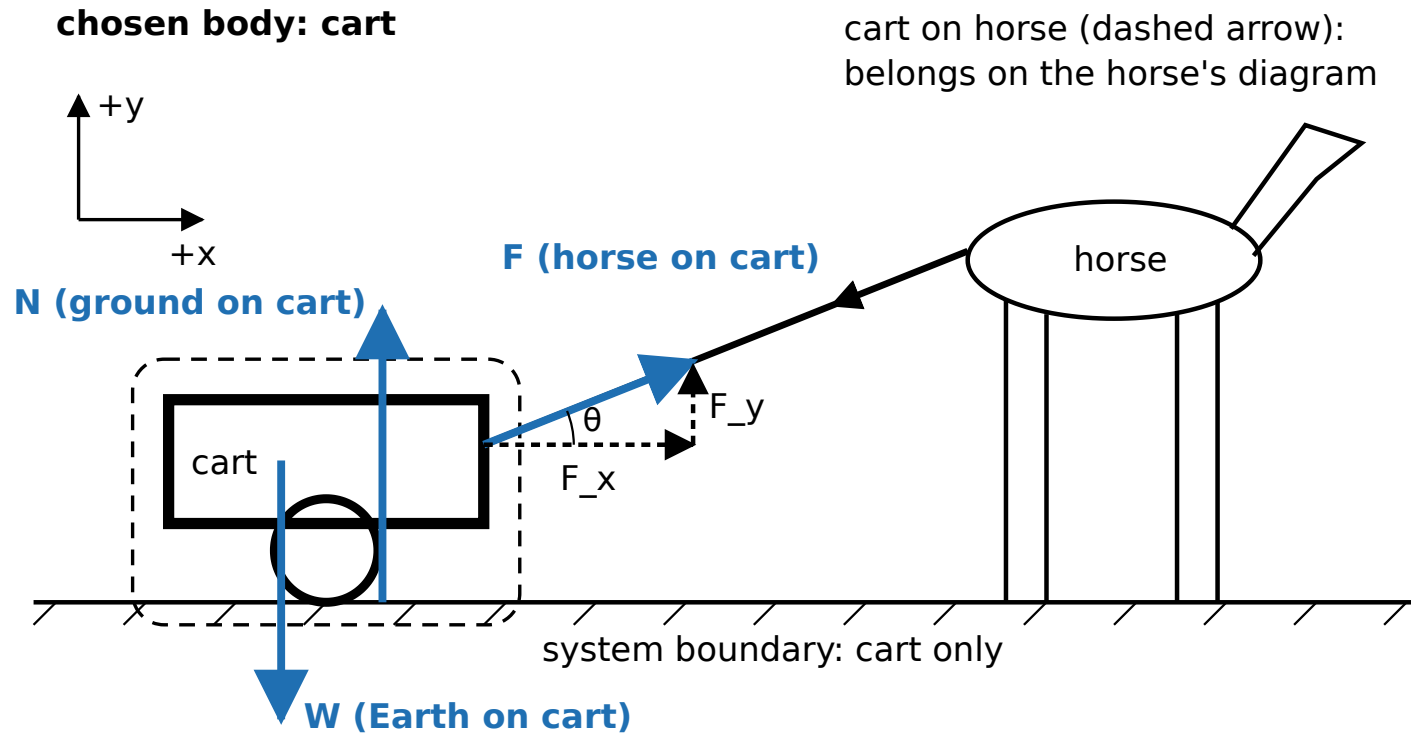
Established: a horizontal push splits into $P \cos \theta$ along the slope and $P \sin \theta$ into it, so both N and the acceleration change. The transfer task adds friction, where that extra normal force raises the friction limit.

Inclined-plane force modelling

A block rests on a smooth fixed incline. A student draws three downward-related force arrows on the same free-body diagram: mg vertically downward, $mg \sin(\theta)$ down the plane, and $mg \cos(\theta)$ into the plane. Diagnose the diagram and state how the weight should be represented when writing plane-aligned equations.

Conditions

- The only physical interactions are Earth-on-block gravity and plane-on-block normal contact.
- The question is about force ownership and representation, not numerical calculation.



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

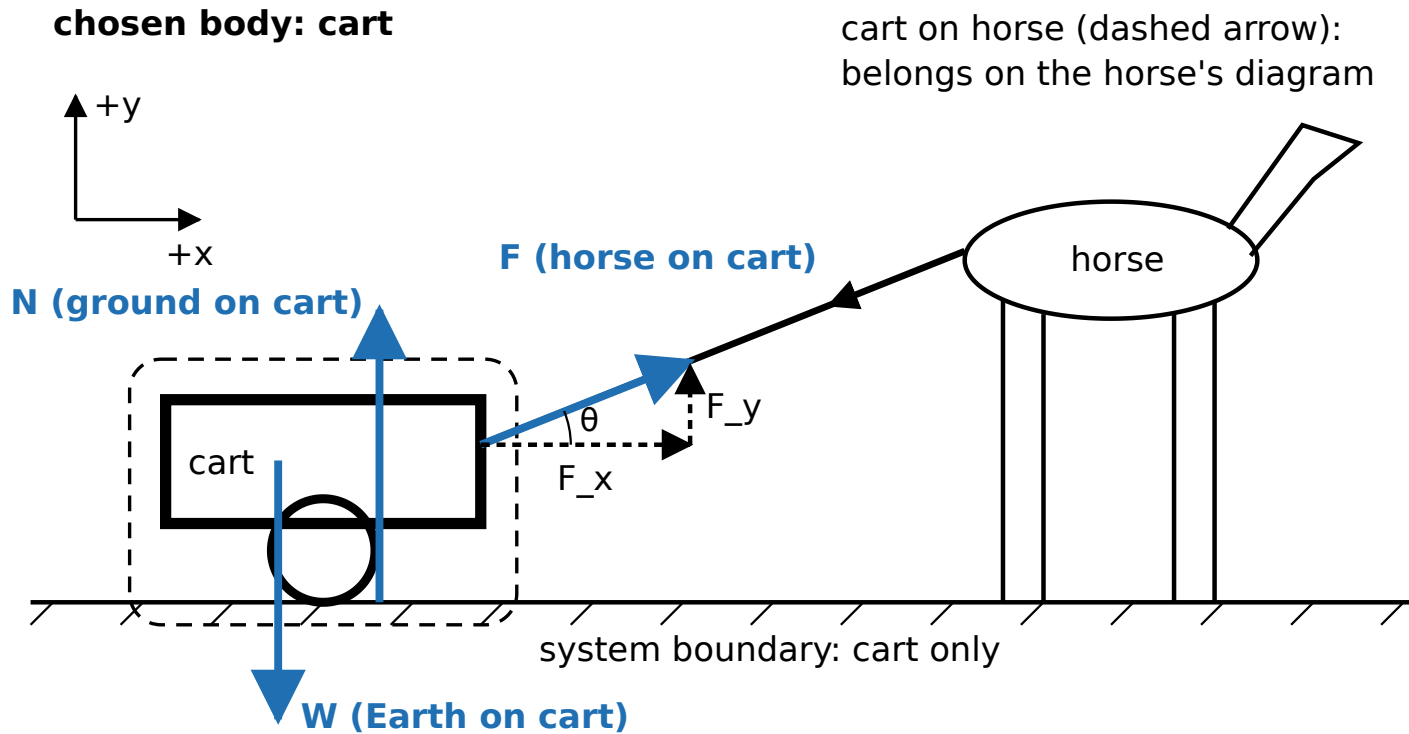
Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. Did choosing new axes create any new physical object acting on the block?

Reasoning route and full solution



External agents acting on the cart: horse, Earth, ground

Every arrow answers "who acts on the cart?". Partner forces act on the other body and never cancel on this diagram.

Components: $F_x = F \cos \theta$ along $+x$, $F_y = F \sin \theta$ along $+y$.

They replace F in the axis equations; they are not extra forces.

Make force ownership visible for one chosen body: identify the body and external agents first, then represent only forces acting on that body; resolved components remain a coordinate representation of one physical force.

1. A free-body inventory is a list of physical interactions acting on the chosen body.
2. Gravity is one Earth-on-block interaction, represented by the vector mg .
3. Resolving that vector into parallel/perpendicular components changes its coordinate description only.
4. The component equations may use the two components instead of the unresolved weight vector; they are not two new agents.

Answer

The diagram double-counts gravity. Weight mg is one physical force. For plane-aligned equations it may be represented by the components $mg \sin(\theta)$ down the plane and $mg \cos(\theta)$ into the plane, but the original mg must not also be added as a separate force in those component sums.

Independent check

The physical-force count remains two in this smooth-contact case: weight and normal reaction.

If you went wrong

You kept mg alongside $mg \sin \theta$ and $mg \cos \theta$: that counts gravity three times; the components replace the weight, they do not join it.

Repair

Revisit: When a physical force is resolved into components for chosen axes, keep the body-force inventory unchanged: use the components as a representation of that one force in the equations, not as additional interaction forces alongside the original vector.

What this establishes

Established: resolving a force changes its description, not the number of forces acting. The transfer task finds the same double count written into component equations.